Oracle 19c database Standalone installation with ASM: install database on ASM File System/ ASM storage – restart server - TESTDB – oraclelab2

Standalone with ASM -> oraclelab2.rar (Link #1) (ASM file system / ASM storage) - TESTDB

oraclelab2.localdomain.com (Linux VM 7.9)

1. Setup oraclelab2 server - 192.168.0.152
2. Download 19c ASM Software and Install and Setup ASM storage – this step is extra if we compare with Standalone without ASM
3. Download 19c Oracle Software and Install
4. Create Database -> TESTDB

1. Setup oraclelab2 server -

Download oraclelab2.rar file from Link #1 >>> Ready_VMs_Standalone (folder)

https://drive.google.com/drive/folders/183dcnQtLtEp8mS3MX3VvCyiKjrpNGicT?usp=sharing



Download and install winrar from link #2 software to unzip oraclelab1.rar >>> Other Tools for DBAs (folder)

https://drive.google.com/drive/folders/1mh5j3hBMad0cCpV7mhph2mmyAGYfwx3U?usp=sharing



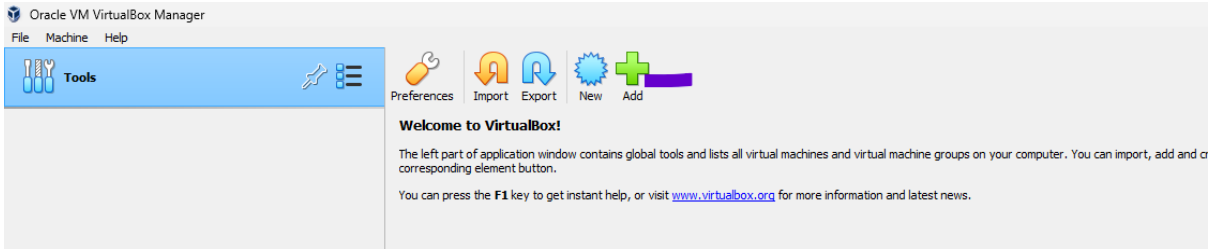
Download and Install Oracle Virtualbox from Link# 2 and setup oraclelab1 >>> Other Tools for DBAs (folder)

https://drive.google.com/drive/folders/1mh5j3hBMad0cCpV7mhph2mmyAGYfwx3U?usp=sharing

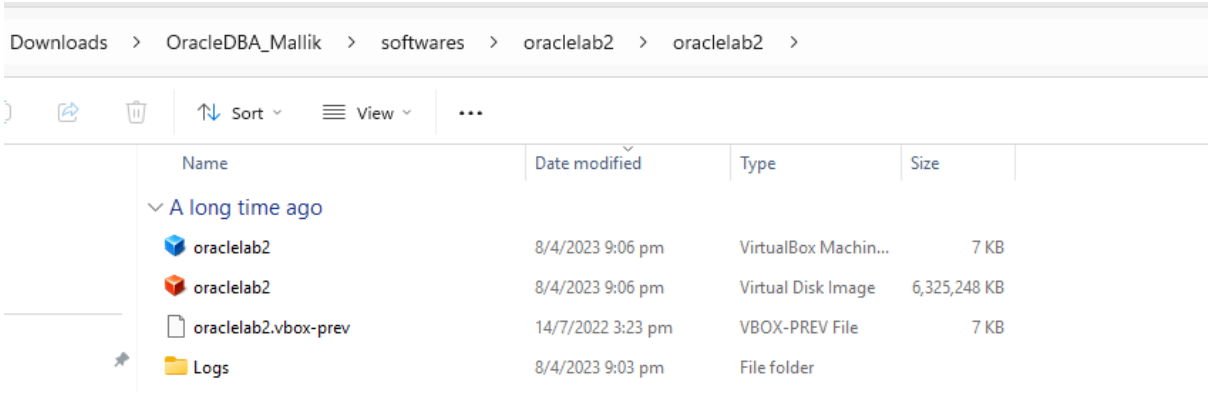


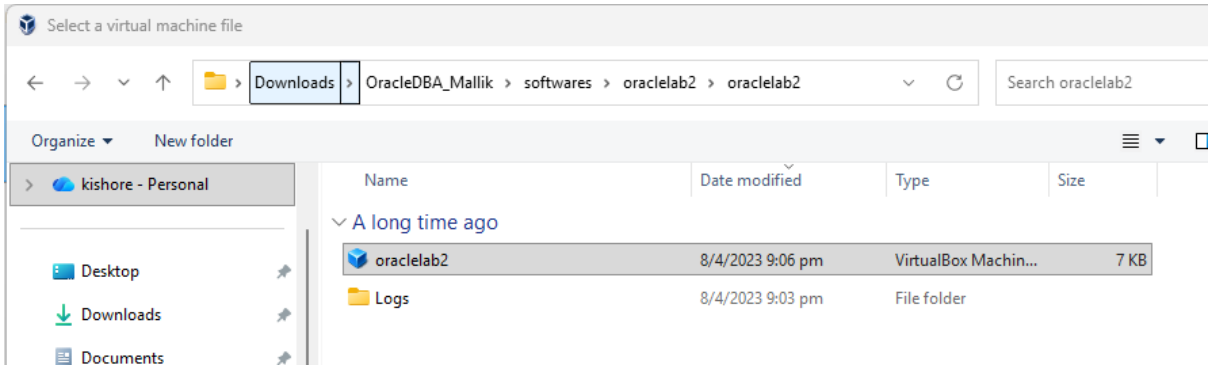
Post installation of VirtualBox, open – add

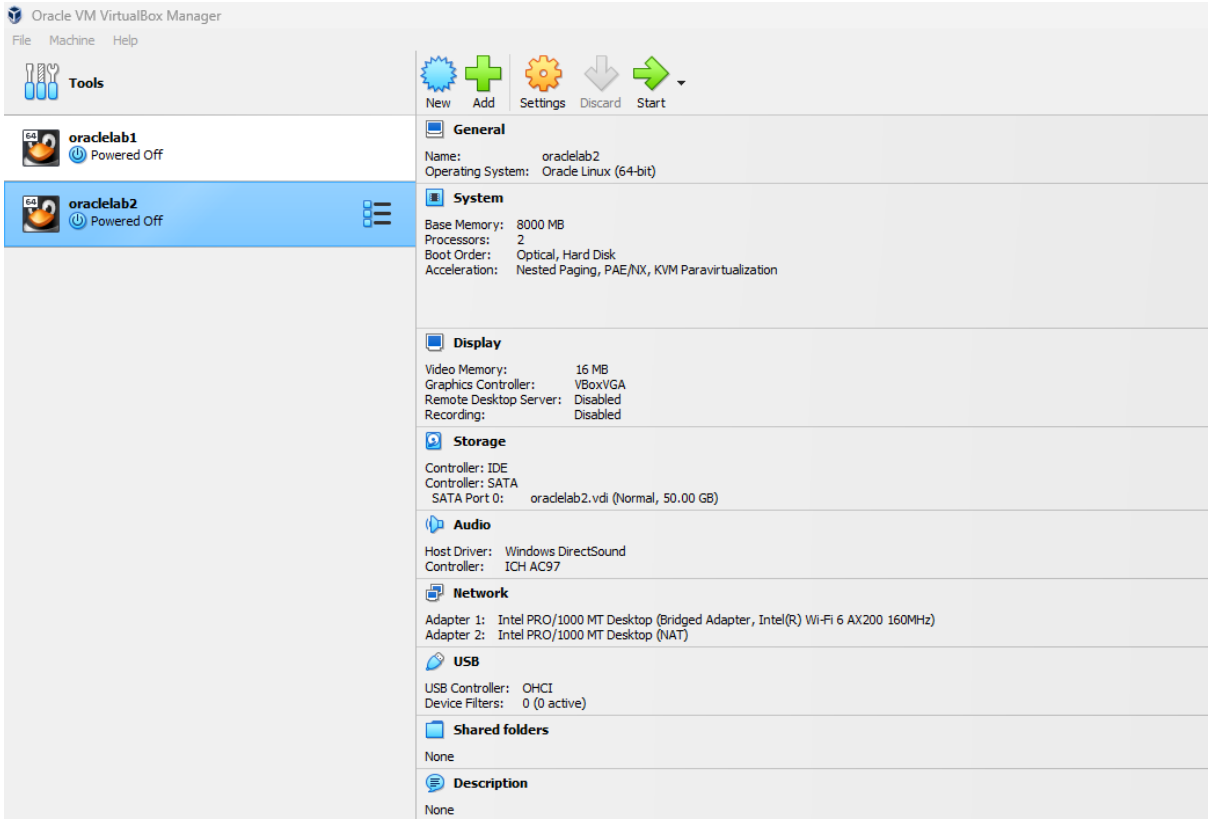
- Setup oraclelab1 - add oraclelab1 to oracle virtual box



After extracting oraclelab2.rar

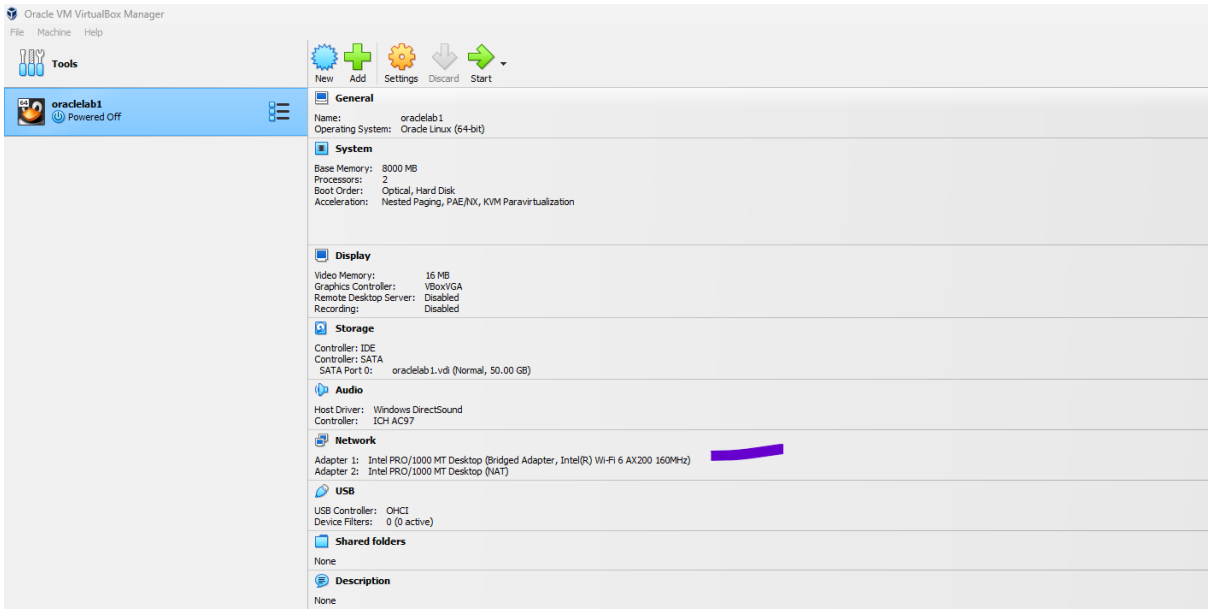






<u>Network setup</u> (Its done by network admin and they will assign IP address)

Remove the existing network settings and add our own laptop IP -



Both make it as NAT

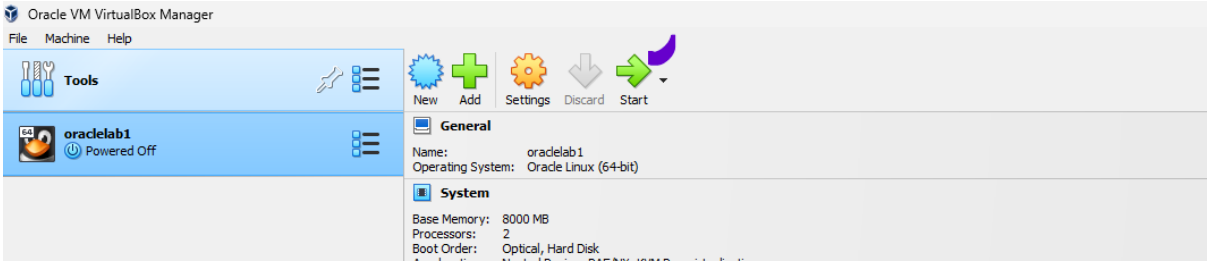


Then again change them back to Bridge adapter

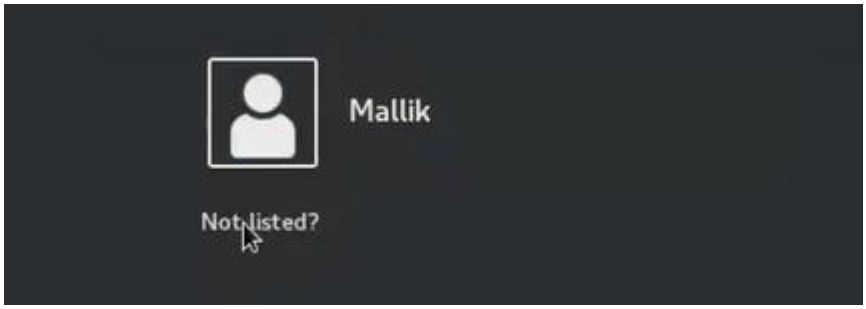


The moment we change to bridged adapter it will pick up the new laptop IP details –

Now, start the VM –



Don't click on Malik user – click on not listed



Root – pwd: Mallik@123

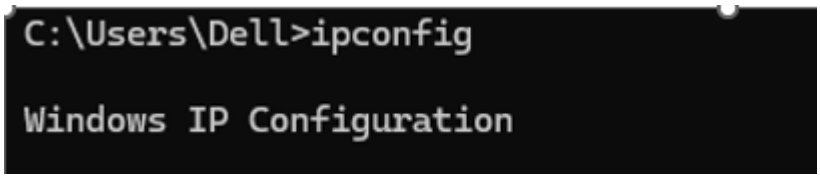


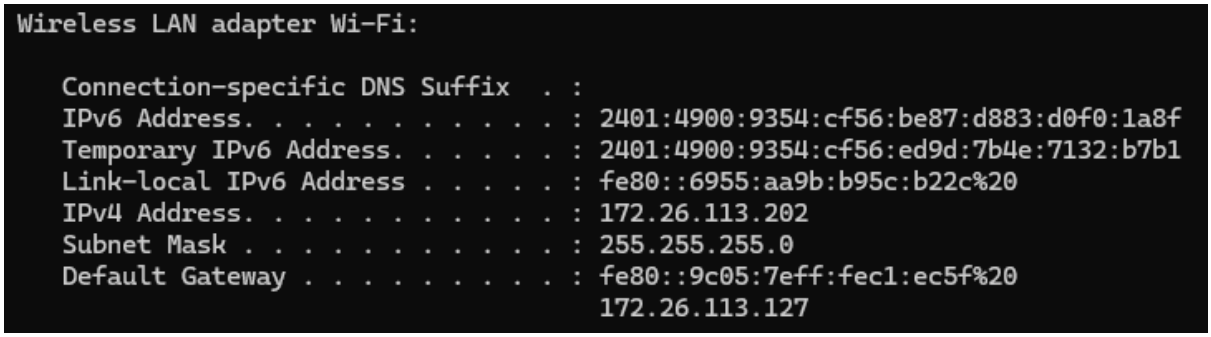
Next, IP configuration –

From local laptop get these details -

cmd> ipconfig

cmd> ipconfig /all



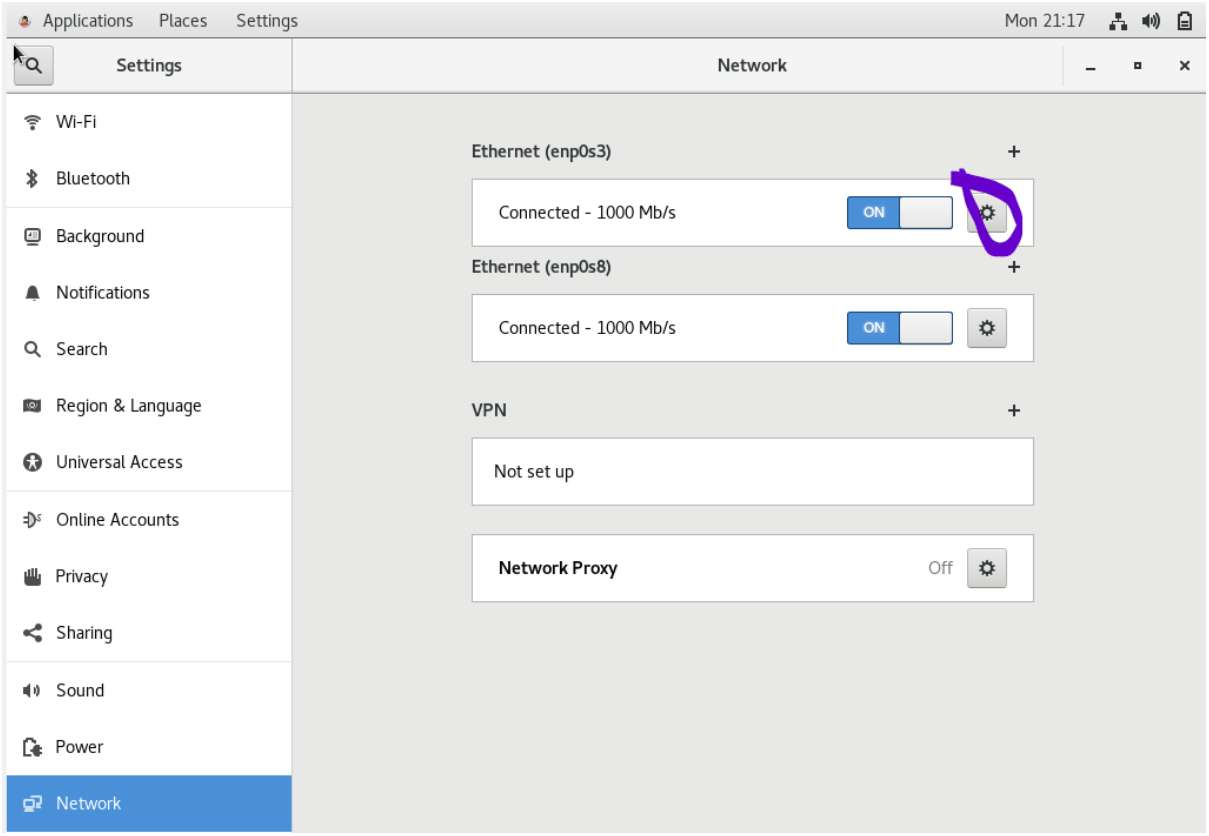


- get laptop/desktop IP address -> 172.26.113.202
- Subnet - 255.255.255.0
- gateway -> 172.26.113.127
- Get DNS IP address from laptop/desktop -> 172.26.113.127

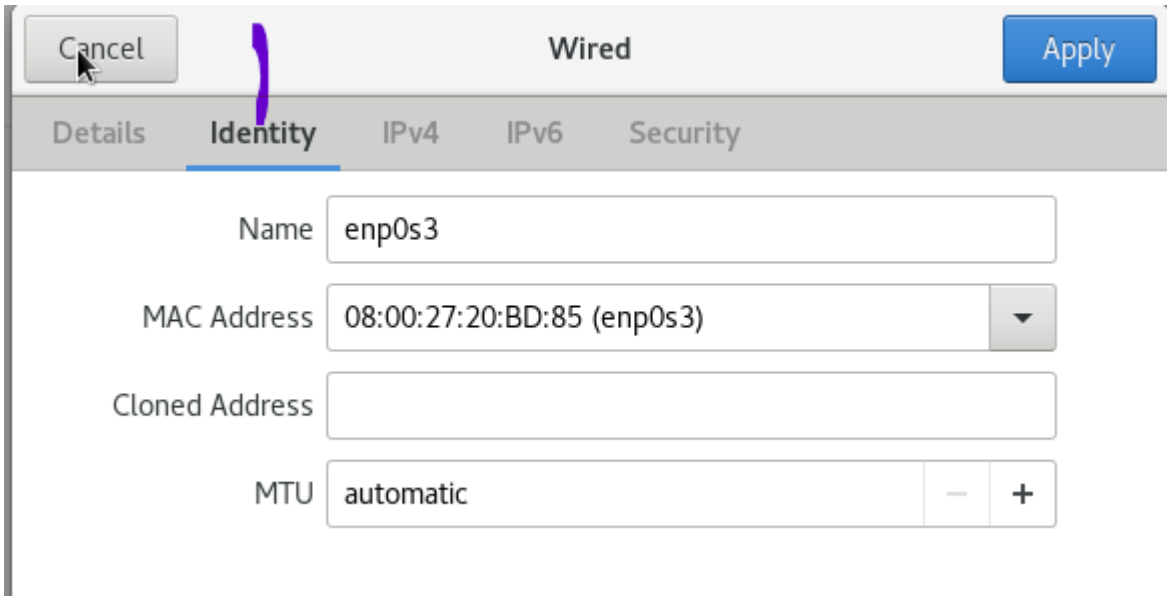
- Assign IP address to VM oraclelab2 in the same range of your laptop/desktop IP address -> **172.26.113.203**

Assign this new IP – VirtualBox

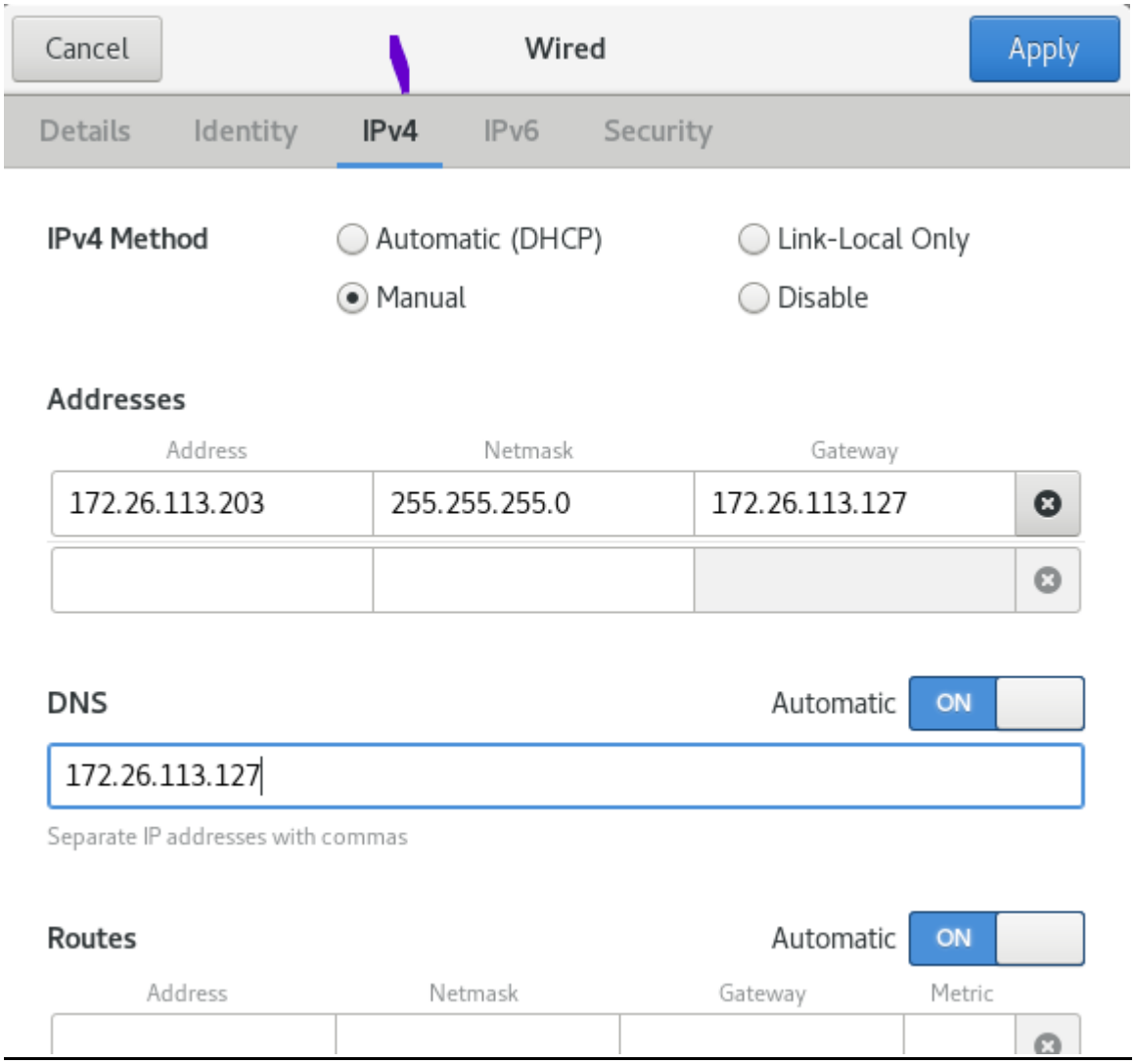
Applications -> system tools -> settings -> Network


Applications
Places
Settings
Mon 21:17
Settings
Network
Wi-Fi
Bluetooth
Background
Notifications
Search
Region & Language
Universal Access
Online Accounts
Privacy
Sharing
Sound
Power
Network
Ethernet (enp0s3)
Connected - 1000 Mb/s
ON
Ethernet (enp0s8)
Connected - 1000 Mb/s
ON
VPN
Not set up
Network Proxy
Off

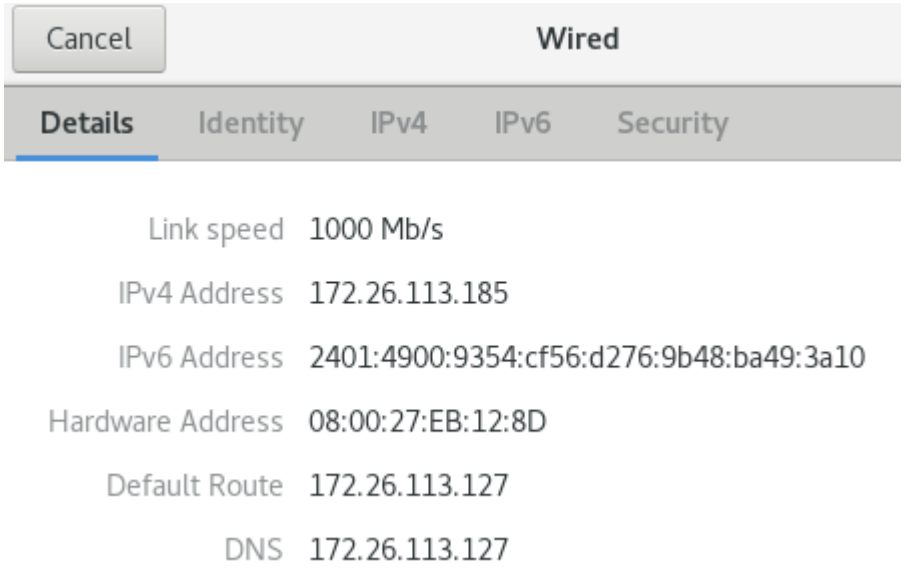
Settings ->


Cancel
Wired
Apply
Details
Identity
IPv4
IPv6
Security
Name
enp0s3
MAC Address
08:00:27:20:BD:85 (enp0s3)
Cloned Address
MTU
automatic

IPV4 tab –



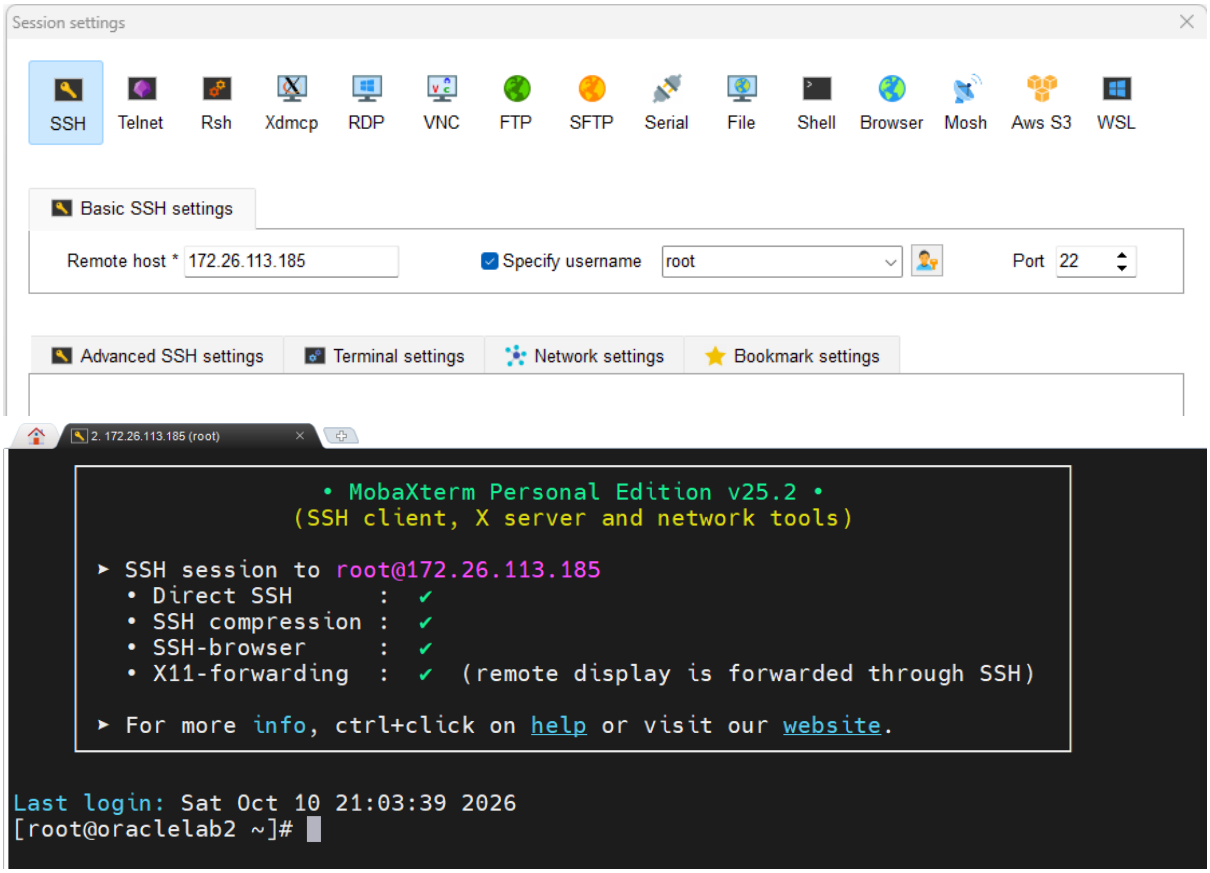
Click apply – then check –



- Access oraclelab2 (172.26.113.185) from putty / mobaxterm, putty / mobaxterm software can be downloaded from link #2 and Install on your laptop >>> Other Tools for DBAs (folder)

https://drive.google.com/drive/folders/1mh5j3hBMad0cCpV7mhph2mmyAGYfwx3U?usp=sharing

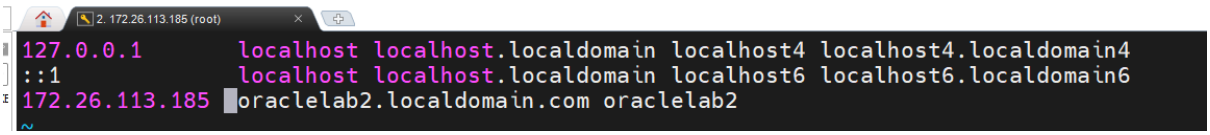
From MobaXterm – connect –



- add IP address (172.26.113.185) into vi /etc/hosts file

172.26.113.185 oraclelab2.localdomain.com oraclelab2





Validate

```
[root@oraclelab2 ~]# hostname
oraclelab2.localdomain.com
```

```
[root@oraclelab2 ~]# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.26.113.185  netmask 255.255.255.0  broadcast 172.26.113.255
        inet6 2401:4900:9354:cf56:d276:9b48:ba49:3a10  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::9826:ec83:cd25:7e55  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:eb:12:8d  txqueuelen 1000  (Ethernet)
        RX packets 2298  bytes 660909 (645.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3126  bytes 394213 (384.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

2. Download 19c ASM Software and Install and Setup ASM storage

GI Home (Grid Infrastructure Home) (or) ASM Home (Automatic Storage Management Home) (or) GI Software (or) ASM Software (or) Grid Home (or) Clusterware Software (or) Clusterware Home – all meanings are same

a. Download ASM/GI software (19c) - https://edelivery.oracle.com (Oracle Database 19c 19.3.0.0.0) (or)

- Oracle Database 19c 19.3.0.0.0 (Oracle Database In-Memory, Oracle Database Vault)

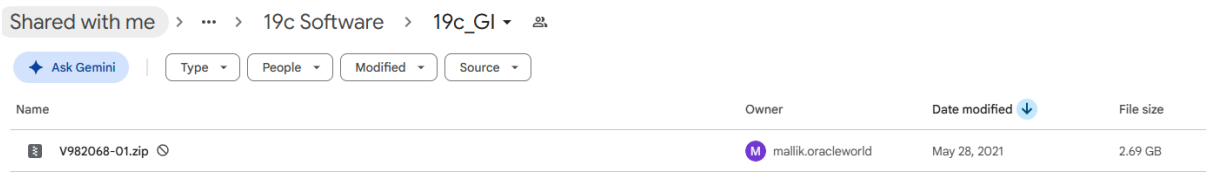
- Download it from Google drive Link #2 >>> Oracle Software and Patches >>> 19c Software

- https://drive.google.com/drive/folders/1mh5j3hBMad0cCpV7mhph2mmyAGYfwx3U

- DB - V982063-01.zip

- GI/ASM - V982068-01.zip



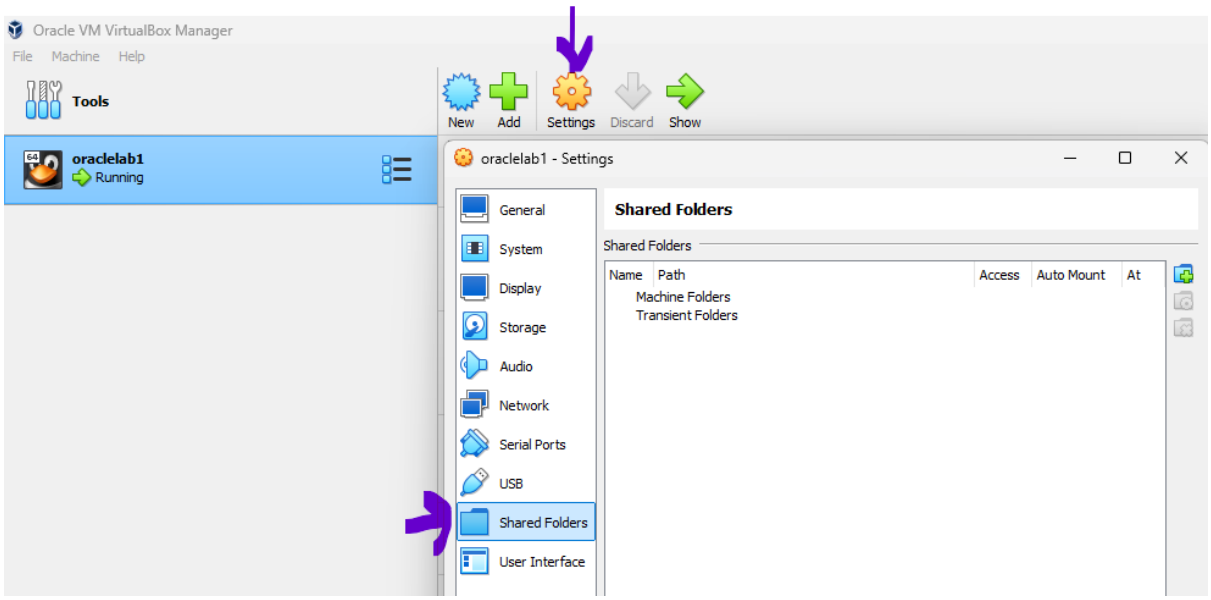


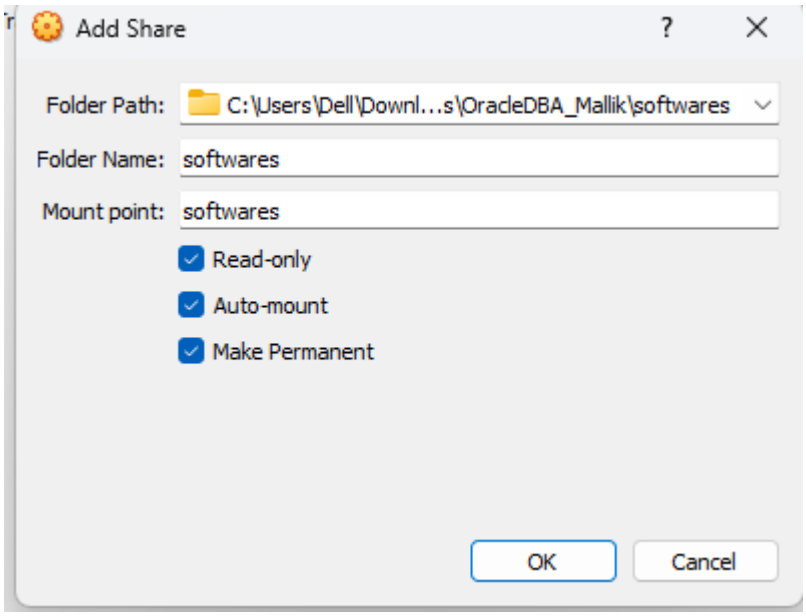
b. Install GI Software / ASM Home / Clusterware Software / ASM Software / GI Home

- Copy GI SW from laptop/Desktop to VM (oraclelab2)

- Use winscp

- **Shared folder (preferred)**





We can see the shared folder in linux



```
root@oraclelab2:/media/sf_softwares
File  Edit  View  Search  Terminal  Help
[root@oraclelab2 ~]# df -h
Filesystem           Size  Used Avail Use% Mounted on
devtmpfs             3.7G     0  3.7G   0% /dev
tmpfs                3.7G     0  3.7G   0% /dev/shm
tmpfs                3.7G  9.4M  3.7G   1% /run
tmpfs                3.7G     0  3.7G   0% /sys/fs/cgroup
/dev/mapper/ol-root   34G  5.7G   29G  17% /
/dev/sda1           1014M  231M  784M  23% /boot
softwares            476G  342G  135G  72% /media/sf_softwares
tmpfs                749M   20K  749M   1% /run/user/0
[root@oraclelab2 ~]# cd /media/sf_softwares/
[root@oraclelab2 sf_softwares]# ls
oraclelab1      V982063-01-002      V982068-01-001.zip
oraclelab1.rar  V982063-01-002.zip  V982068-01.zip
oraclelab2      V982068-01          VirtualBox-7.0.20-163906-Win.exe
oraclelab2.rar  V982068-01-001      winrar-x64-621.exe
[root@oraclelab2 sf_softwares]#
```

- Directory creation for grid home and dbhome & SW unzip

- mkdir -p /u01/app/19.0.0.0/grid

- unzip /media/sf_Software/Grid_19c_Software/V982068-01.zip -d /u01/app/19.0.0.0/grid (or)

- **unzip V982068-01.zip -d /u01/app/19.0.0.0/grid**

```
/u01/app/19.0.0.0/grid/lib/libclntsh.so.10.1 -> libclntsh.so
/u01/app/19.0.0.0/grid/jdk/jre/bin/ControlPanel -> jcontrol
/u01/app/19.0.0.0/grid/javavm/admin/libjtcjt.so -> ../../javavm/jdk/jdk8/admin/libjtcjt.so
/u01/app/19.0.0.0/grid/javavm/admin/classes.bin -> ../../javavm/jdk/jdk8/admin/classes.bin
/u01/app/19.0.0.0/grid/javavm/admin/lfclasses.bin -> ../../javavm/jdk/jdk8/admin/lfclasses.bin
/u01/app/19.0.0.0/grid/javavm/lib/security/cacerts -> ../../../javavm/jdk/jdk8/lib/security/cacerts
/u01/app/19.0.0.0/grid/javavm/lib/sunjce_provider.jar -> ../../javavm/jdk/jdk8/lib/sunjce_provider.jar
/u01/app/19.0.0.0/grid/javavm/lib/security/README.txt -> ../../../javavm/jdk/jdk8/lib/security/README.txt
/u01/app/19.0.0.0/grid/javavm/lib/security/java.security -> ../../../javavm/jdk/jdk8/lib/security/java.security
/u01/app/19.0.0.0/grid/jdk/jre/lib/amd64/server/libjsig.so -> ../libjsig.so
[root@oraclelab2 sf_softwares]#
```

- mkdir -p /u01/app/oracle/product/19.0.0.0/dbhome_1

- unzip /media/sf_Software/DB_19c_Software/V982063-01.zip -d /u01/app/oracle/product/19.0.0.0/dbhome_1 (or)

- **unzip V982063-01.zip -d /u01/app/oracle/product/19.0.0.0/dbhome_1**

```
[root@oraclelab2 sf_softwares]# mkdir -p /u01/app/oracle/product/19.0.0.0/dbhome_1
[root@oraclelab2 sf_softwares]# unzip V982063-01.zip -d /u01/app/oracle/product/19.0.0.0/dbhome_1
unzip:  cannot find or open V982063-01.zip, V982063-01.zip.zip or V982063-01.zip.ZIP.
[root@oraclelab2 sf_softwares]# unzip V982063-01-002.zip -d /u01/app/oracle/product/19.0.0.0/dbhome_1/
```

```
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/admin/lfclasses.bin -> ../../javavm/jdk/jdk8/admin/lfclasses.bi
n
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/lib/security/cacerts -> ../../../javavm/jdk/jdk8/lib/security/c
acerts
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/lib/sunjce_provider.jar -> ../../javavm/jdk/jdk8/lib/sunjce_pro
vider.jar
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/lib/security/README.txt -> ../../../javavm/jdk/jdk8/lib/securit
y/README.txt
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/lib/security/java.security -> ../../../javavm/jdk/jdk8/lib/secu
rity/java.security
  /u01/app/oracle/product/19.0.0.0/dbhome_1/jdk/jre/lib/amd64/server/libjsig.so -> ../libjsig.so
[root@oraclelab2 sf_softwares]#
```

id oracle -> no oracle user – after rpm installation oracle user will be created

```
[root@oraclelab2 DB]# id oracle
id: oracle: no such user
[root@oraclelab2 DB]# yum install oracle* --skip-broken
Loaded plugins: langpacks, ulninfo
Package oracle-logos-70.7.0-1.0.7.el7.noarch already installed and latest version
Package oracle-spacewalk-client-release-el7 is obsoleted by oracle-linux-manager-
```

- **yum install oracle* --skip-broken**

    - It will set all kernel parameters and
    - pre-requisite rpms and
    - shared memory setting
    - auto create users and groups

- yum -y install oracle-database-ee-19c
- yum -y install oracle-database-ee-21c

    https://docs.oracle.com/en/database/oracle/oracle-database/19

- Optional (In case if you installed above RPM) (Not Needed)
- User & group create

    # groupadd dba

    # groupadd oinstall

    # useradd -g oinstall -G oinstall,dba oracle

Advantages of oracle pre-requisite rpm installation

- all kernel parameter are set

- all the required rpms are auto installed

- all the oracle user and groups are auto created

```
[root@oraclelab2 sf_softwares]# yum install oracle* --skip-broken
Loaded plugins: langpacks, ulninfo
No package oraclelab1 available.
No package oraclelab1.rar available.
No package oraclelab2 available.
No package oraclelab2.rar available.
Error: Nothing to do
```

```
root@oraclelab2:/u01/app/oracle/product/19.0.0.0/dbhome_1
File  Edit  View  Search  Terminal  Help
[root@oraclelab2 sf_softwares]# cd /u01/app/oracle/product/19.0.0.0/dbhome_1/
[root@oraclelab2 dbhome_1]# ls
addnode     cv           drdaas         javavm  network  ords     R              schagent.conf  suptools
apex        data         dv             jdbc    nls      oss      racg           sdk            ucp
assistants  dbjava       env.ora        jdk     odbc     oui      rdbms          slax           usm
bin         dbs          has            jlib    olap     owm      relnotes       sqldeveloper   utl
clone       deinstall    hs             ldap    OPatch   perl     root.sh        sqlj           wwg
crs         demo         install        lib     opmn     plsql    root.sh.old    sqlpatch       xdk
css         diagnostics  instantclient  md      oracore  precomp  root.sh.old.1  sqlplus
ctx         dmu          inventory      mgw     ord      QOpatch  runInstaller   srvm
[root@oraclelab2 dbhome_1]# yum install oracle* --skip-broken
```

```
  oraclelinux-developer-release-el7.x86_64 0:1.0-6.el7

Dependency Installed:
  ksh.x86_64 0:20120801-144.0.1.el7_9                    libaio-devel.x86_64 0:0.3.109-13.el7

Updated:
  oraclelinux-release.x86_64 7:7.9-1.0.13.el7             oraclelinux-release-el7.x86_64 0:1.0-17.el7

Skipped (dependency problems):
  oracle-linux-manager-client-release-el7.x86_64 0:1.0-2.el7

Complete!
[root@oraclelab2 dbhome 1]#
```

Oracle user is created

```
[root@oraclelab2 dbhome_1]# id oracle
uid=54321(oracle) gid=54321(oinstall) groups=54321(oinstall),54322(dba)
[root@oraclelab2 dbhome_1]#
```

- Change permission

#chown -R oracle:oinstall /u01

```
[root@oraclelab2 dbhome_1]# chown -R oracle:oinstall /u01
[root@oraclelab2 dbhome 1]#
```

- passwd oracle (set password)

```
[root@oraclelab2 dbhome_1]# passwd oracle
Changing password for user oracle.
New password:
Retype new password:
passwd: all authentication tokens updated succe
[root@oraclelab2 dbhome 1]#
```

Password: Mallik@123

Our lab (installed with oracle user):

========

ASM - installed with oracle user

DB - installed with oracle user

========in some clients they will install with below users

ASM - installed with grid user

DB - installed with oracle user

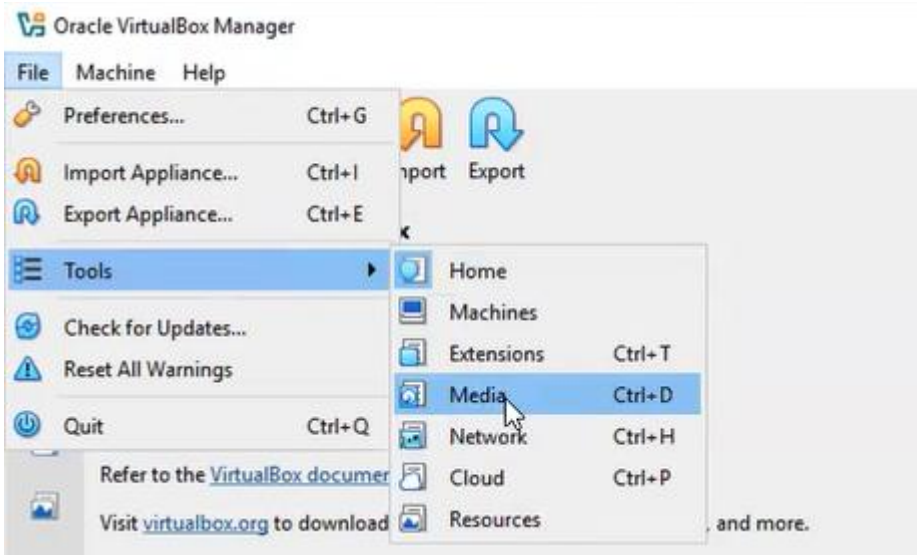
========in some clients' custom installation as below

ASM - installed with oraasm/oragrid/grid user

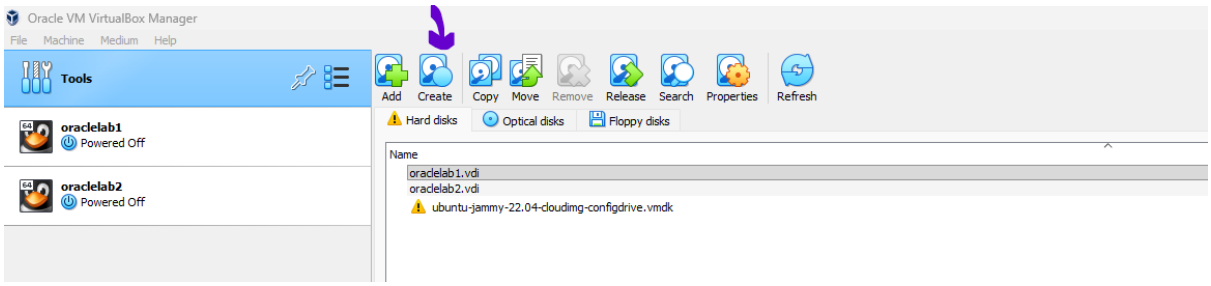
DB - installed with oraprod/orauat/oradev user

c. shared disks / raw disks for ASM installation

- add 3 virtual disks to VM (oraclelab2) - /dev/sdb, /dev/sdc, /dev/sdd - There are virtual disks (no physical disk for ASM)
- ASM will not understand virtual disks
- ASM will understand only physical disks / raw disks / block devices
- We need to make my ASM fool
    - Tell ASM that these physical disks (/dev/sdb, /dev/sdc, /dev/sdd) by labeling
- We can perform labeling by using oracle asm library



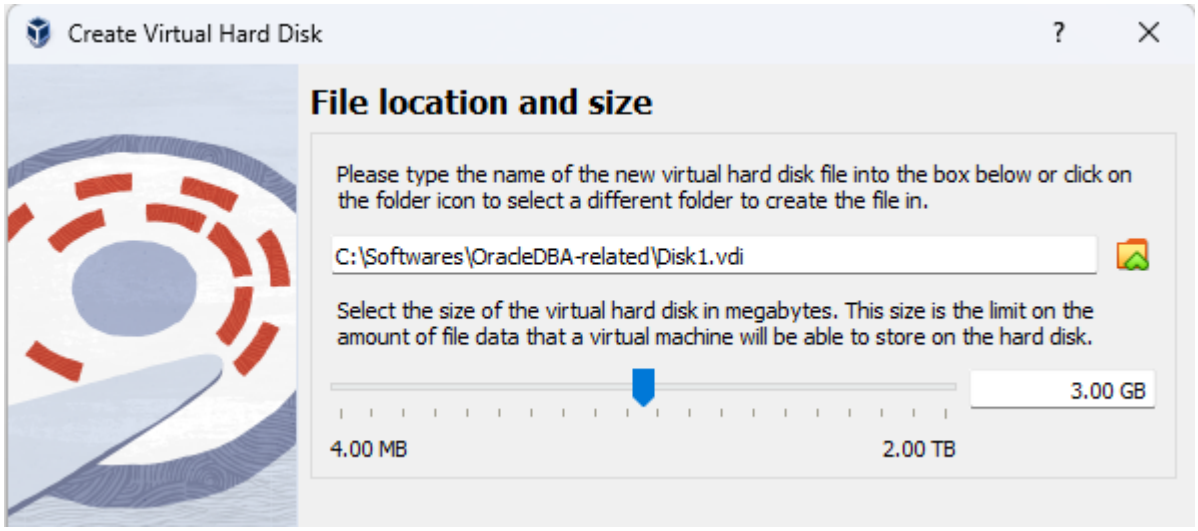
We need to create 3 virtual disks – sdb, sdc, sdd

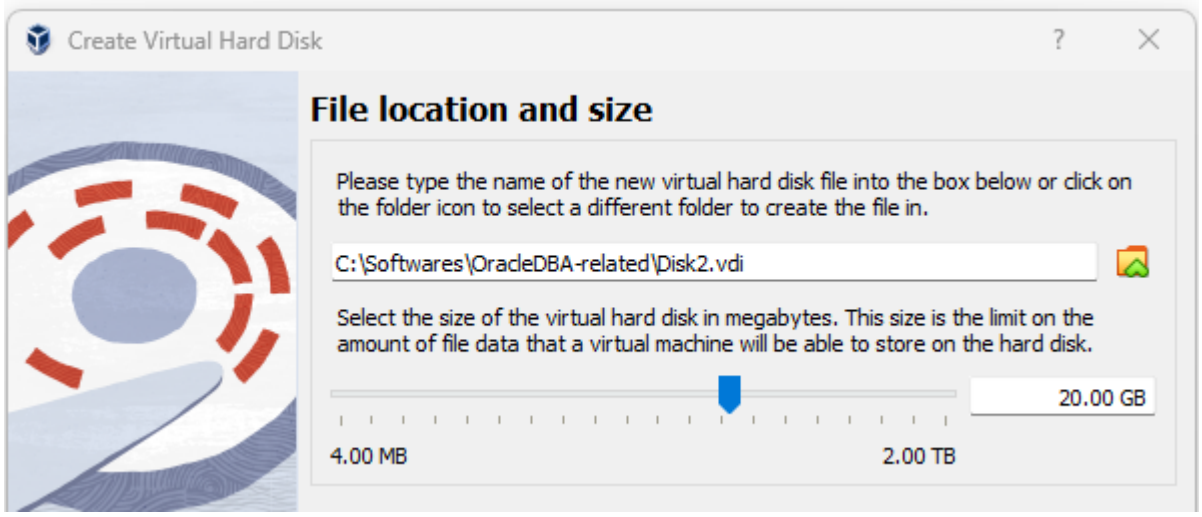


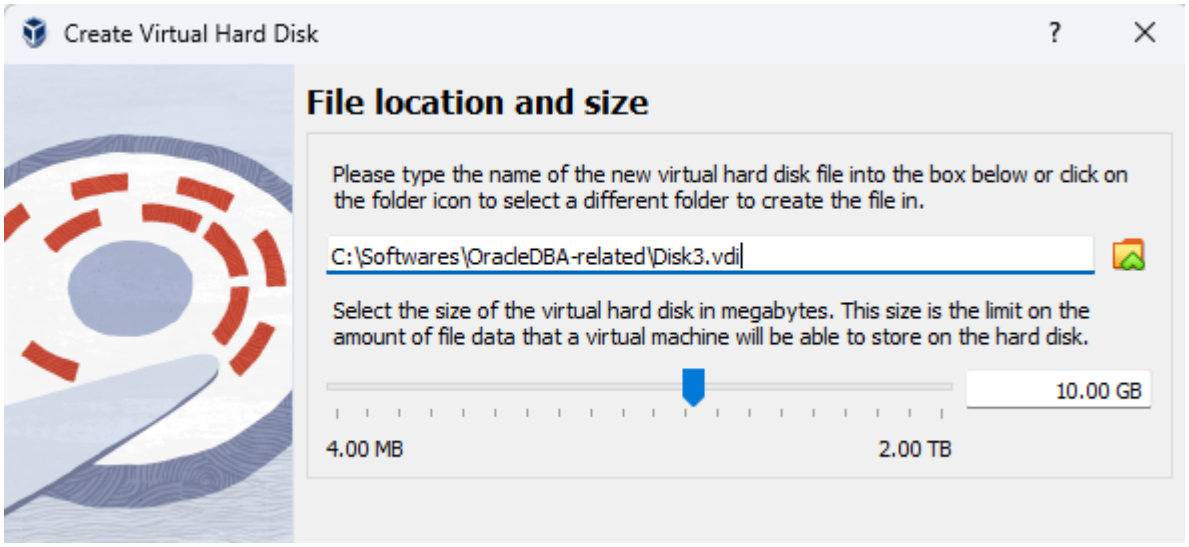


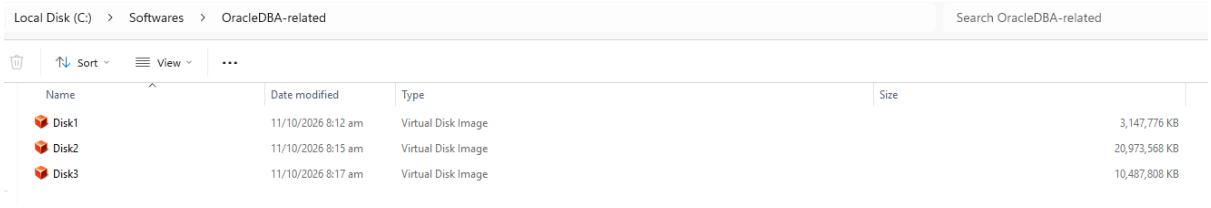
Check the pre-allocate full size

3GB - ASM (ASM related files), 20GB – Database, 10GB - Database

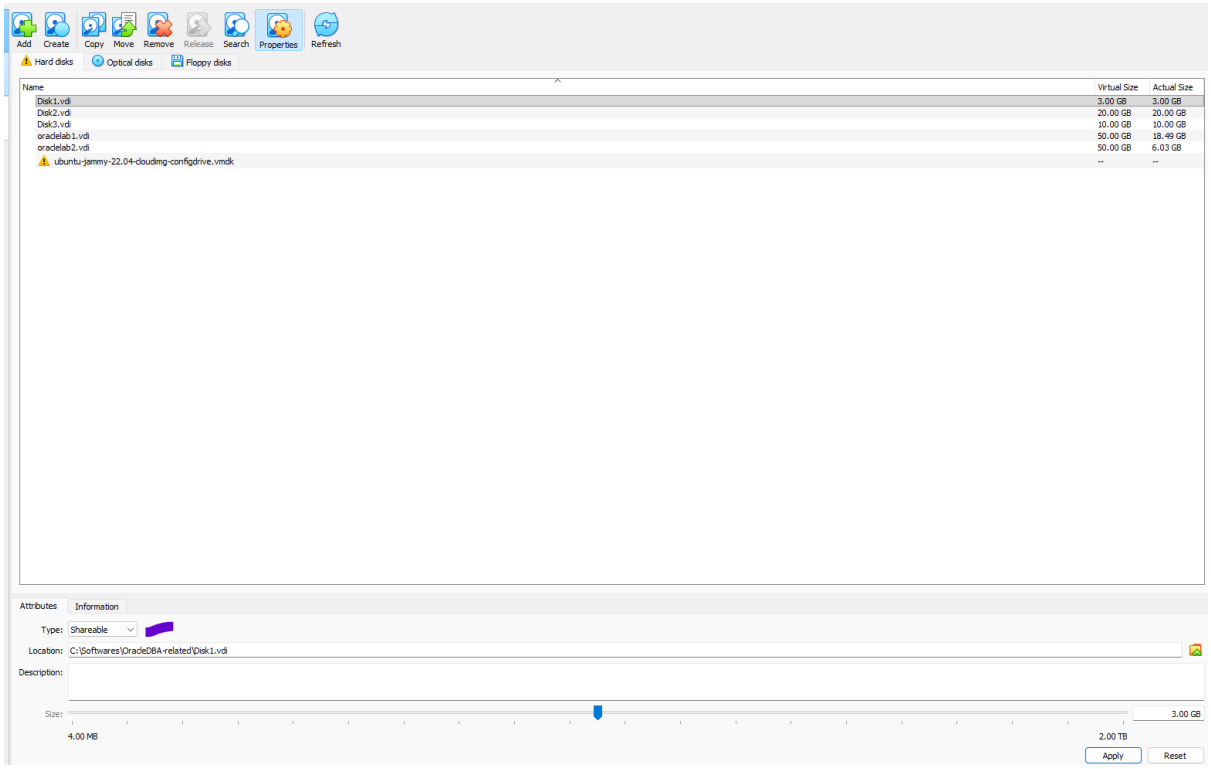




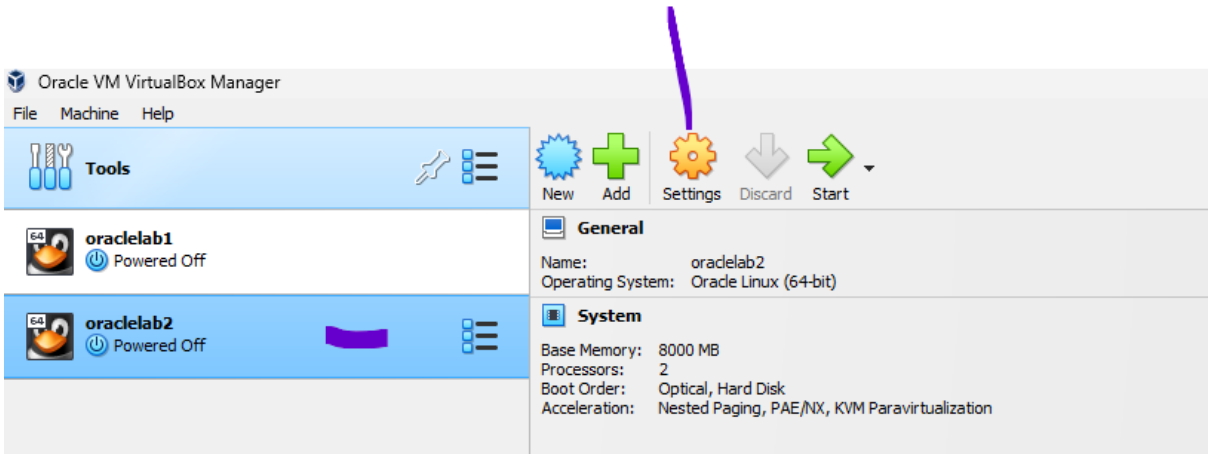




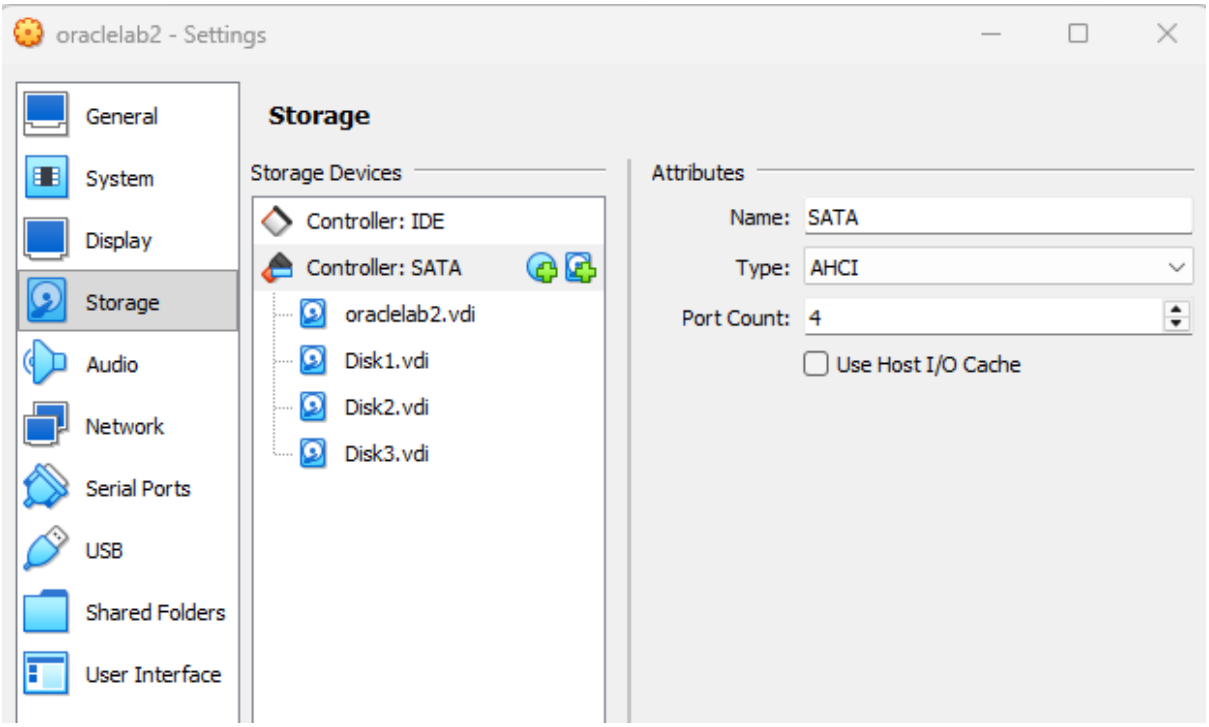
Select Disk1 – Shareable – apply it, same for Disk2, Disk3



Lab2



Settings -> Storage -> Controller SATA -> add Disk1,2,3



OK -> Start OracleLab2 VM -> 3 disks added

```
[root@oraclelab2 ~]# cd /dev
[root@oraclelab2 dev]# ll sd*
brw-rw----. 1 root disk 8,  0 Oct 11 09:36 sda
brw-rw----. 1 root disk 8,  1 Oct 11 09:36 sda1
brw-rw----. 1 root disk 8,  2 Oct 11 09:36 sda2
brw-rw----. 1 root disk 8, 16 Oct 11 09:36 sdb
brw-rw----. 1 root disk 8, 32 Oct 11 09:36 sdc
brw-rw----. 1 root disk 8, 48 Oct 11 09:36 sdd
[root@oraclelab2 dev]#
```

Using fdisk – we can partition them

```
[root@oraclelab2 dev]# fdisk /dev/sdb
Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0x471138af.

Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p):
Using default response p
Partition number (1-4, default 1):
First sector (2048-6291455, default 2048):
Using default value 2048
Last sector, +sectors or +size{K,M,G} (2048-6291455, default 6291455):
Using default value 6291455
Partition 1 of type Linux and of size 3 GiB is set

Command (m for help): w
The partition table has been altered!

Calling ioctl() to re-read partition table.
Syncing disks.
[root@oraclelab2 dev]#
```

```
[root@oraclelab2 dev]# ll sd*
brw-rw----. 1 root disk 8,  0 Oct 11 09:36 sda
brw-rw----. 1 root disk 8,  1 Oct 11 09:36 sda1
brw-rw----. 1 root disk 8,  2 Oct 11 09:36 sda2
brw-rw----. 1 root disk 8, 16 Oct 11 09:48 sdb
brw-rw----. 1 root disk 8, 17 Oct 11 09:48 sdb1
brw-rw----. 1 root disk 8, 32 Oct 11 09:36 sdc
brw-rw----. 1 root disk 8, 48 Oct 11 09:36 sdd
[root@oraclelab2 dev]#
```

So, sdb is partitioned as sdb1

```
[root@oraclelab2 dev]# fdisk /dev/sdc
Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0x5b571aee.

Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p):
Using default response p
Partition number (1-4, default 1):
First sector (2048-41943039, default 2048):
Using default value 2048
Last sector, +sectors or +size{K,M,G} (2048-41943039, default 41943039):
Using default value 41943039
Partition 1 of type Linux and of size 20 GiB is set

Command (m for help): w
The partition table has been altered!

Calling ioctl() to re-read partition table.
Syncing disks.
[root@oraclelab2 dev]#
```

```
[root@oraclelab2 dev]# fdisk /dev/sdd
Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0x9ad5fd77.

Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p):
Using default response p
Partition number (1-4, default 1):
First sector (2048-20971519, default 2048):
Using default value 2048
Last sector, +sectors or +size{K,M,G} (2048-20971519, default 20971519):
Using default value 20971519
Partition 1 of type Linux and of size 10 GiB is set

Command (m for help): w
The partition table has been altered!

Calling ioctl() to re-read partition table.
Syncing disks.
[root@oraclelab2 dev]#
```

```
[root@oraclelab2 dev]# ll sd*
brw-rw----. 1 root disk 8,  0 Oct 11 09:36 sda
brw-rw----. 1 root disk 8,  1 Oct 11 09:36 sda1
brw-rw----. 1 root disk 8,  2 Oct 11 09:36 sda2
brw-rw----. 1 root disk 8, 16 Oct 11 09:48 sdb
brw-rw----. 1 root disk 8, 17 Oct 11 09:48 sdb1
brw-rw----. 1 root disk 8, 32 Oct 11 09:50 sdc
brw-rw----. 1 root disk 8, 33 Oct 11 09:50 sdc1
brw-rw----. 1 root disk 8, 48 Oct 11 09:51 sdd
brw-rw----. 1 root disk 8, 49 Oct 11 09:51 sdd1
[root@oraclelab2 dev]#
```

So, after partition sdb partitioned as sdb1, similarly - sdc1, sdd1

/dev/sdb - 3G -> /dev/sdb1 (fdisk) (For ASM)

/dev/sdc - 20G -> /dev/sdc1 (fdisk) (For Database - Datafiles)

/dev/sdd - 10G -> /dev/sdd1 (fdisk) (For Database - FRA - archive/redo logs/controlfile)

We have to label – on top of partitioned disks we need to label ASM disks

```
oracleasm library to convert or label partitioned disks as RAW / physical disks to make my ASM fool
oracleasm configure -i
oracleasm init
```

```
[root@oraclelab2 dev]# oracleasm configure -i
Configuring the Oracle ASM library driver.

This will configure the on-boot properties of the Oracle ASM library
driver.  The following questions will determine whether the driver is
loaded on boot and what permissions it will have.  The current values
will be shown in brackets ('[]').  Hitting <ENTER> without typing an
answer will keep that current value.  Ctrl-C will abort.

Default user to own the driver interface []: oracle
Default group to own the driver interface []: oinstall
Start Oracle ASM library driver on boot (y/n) [n]: y
Scan for Oracle ASM disks on boot (y/n) [y]: y
Writing Oracle ASM library driver configuration: done
[root@oraclelab2 dev]# oraacleasm init
bash: oraacleasm: command not found...
Similar command is: 'oracleasm'
[root@oraclelab2 dev]# oracleasm init
Creating /dev/oracleasm mount point: /dev/oracleasm
Loading module "oracleasm": oracleasm
Configuring "oracleasm" to use device physical block size
Mounting ASMlib driver filesystem: /dev/oracleasm
[root@oraclelab2 dev]#
```

Using oracleasm libraries, we can configure ASM disks

/dev/sdb - 3G -> /dev/sdb1 (fdisk) -> ASMDISK1 (oracleasm)

/dev/sdc - 20G -> /dev/sdc1 (fdisk) -> ASMDISK2 (oracleasm)

/dev/sdd - 10G -> /dev/sdd1 (fdisk) -> ASMDISK3 (oracleasm)

oracleasm createdisk ASMDISK1 /dev/sdb1

oracleasm createdisk ASMDISK2 /dev/sdc1

oracleasm createdisk ASMDISK3 /dev/sdd1

```
[root@oraclelab2 dev]# oracleasm createdisk ASMDISK1 /dev/sdb1
Writing disk header: done
Instantiating disk: done
[root@oraclelab2 dev]# oracleasm createdisk ASMDISK2 /dev/sdc1
Writing disk header: done
Instantiating disk: done
[root@oraclelab2 dev]# oracleasm createdisk ASMDISK3 /dev/sdd1
Writing disk header: done
Instantiating disk: done
[root@oraclelab2 dev]#
```

[root@oraclelab2 dev]# oracleasm listdisks

```
[root@oraclelab2 dev]# oracleasm listdisks
ASMDISK1
ASMDISK2
ASMDISK3
[root@oraclelab2 dev]#
```

[root@oraclelab2 dev]# cd /dev/oracleasm/disks/

```
[root@oraclelab2 dev]# cd /dev/oracleasm/disks/
[root@oraclelab2 disks]# ls -ltr
total 0
brw-rw----. 1 oracle oinstall 8, 17 Oct 11 09:59 ASMDISK1
brw-rw----. 1 oracle oinstall 8, 33 Oct 11 09:59 ASMDISK2
brw-rw----. 1 oracle oinstall 8, 49 Oct 11 10:00 ASMDISK3
[root@oraclelab2 disks]#
```

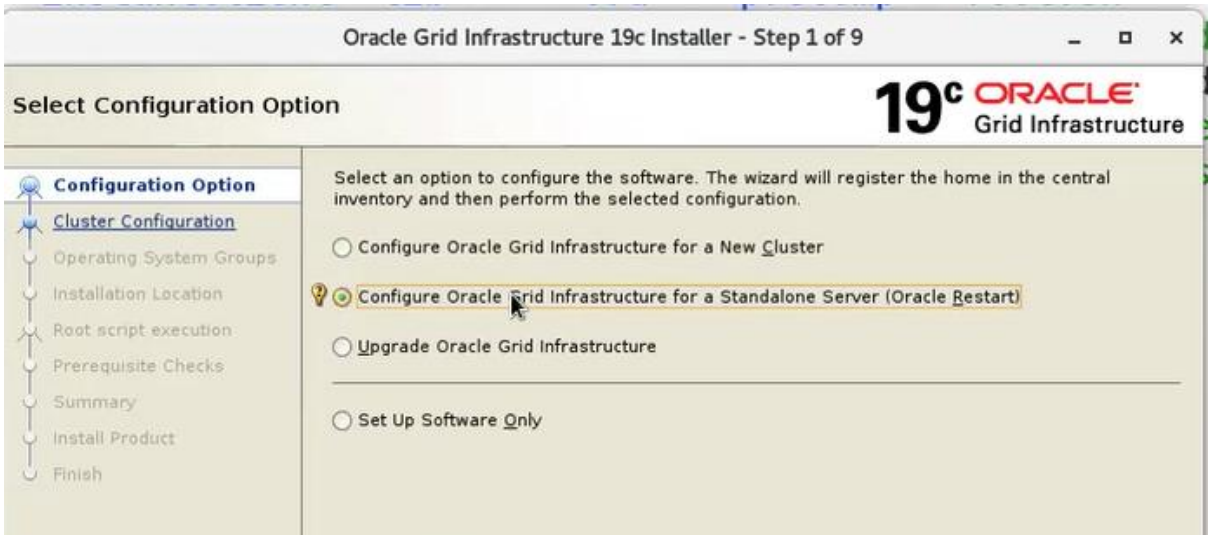
Login with Oracle user -

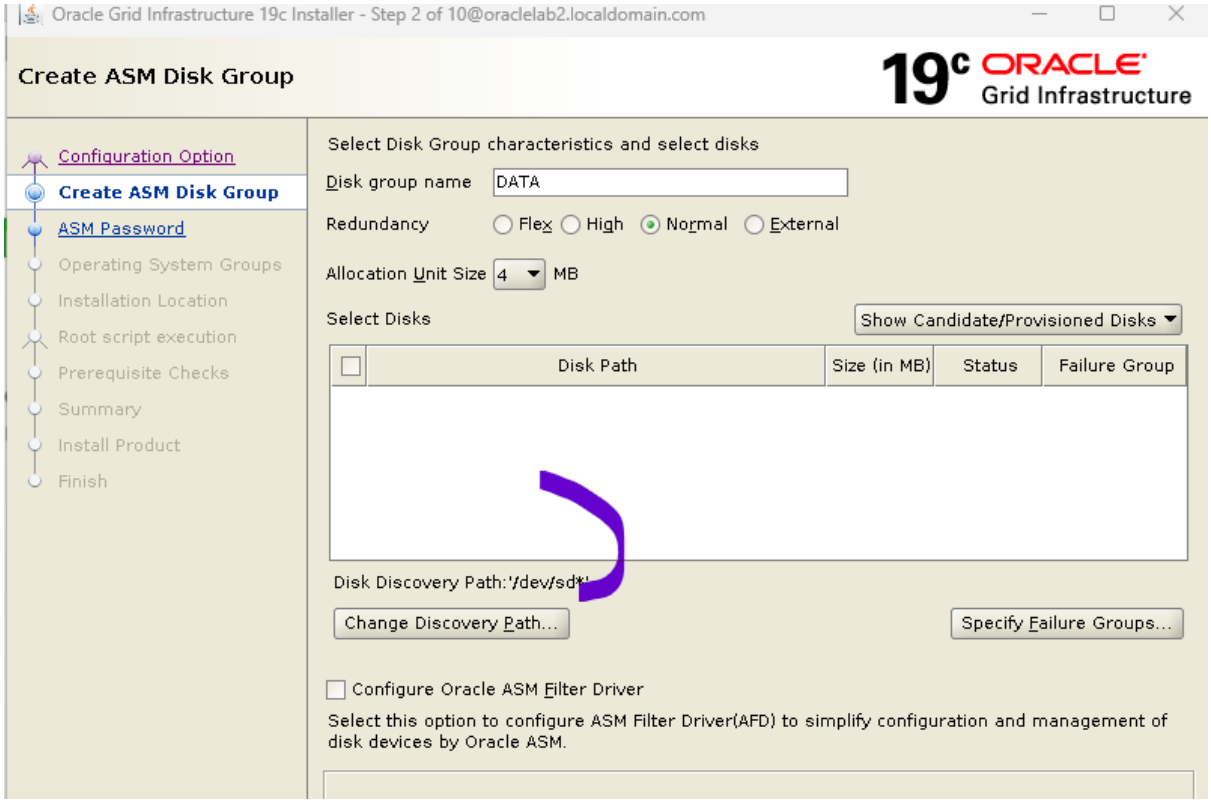
- Now, ASM SW Installation / GI SW Installation / Clusterware Installation (from GUI screen as oracle user)

$ cd /u01/app/19.0.0.0/grid

$ ./gridSetup.sh



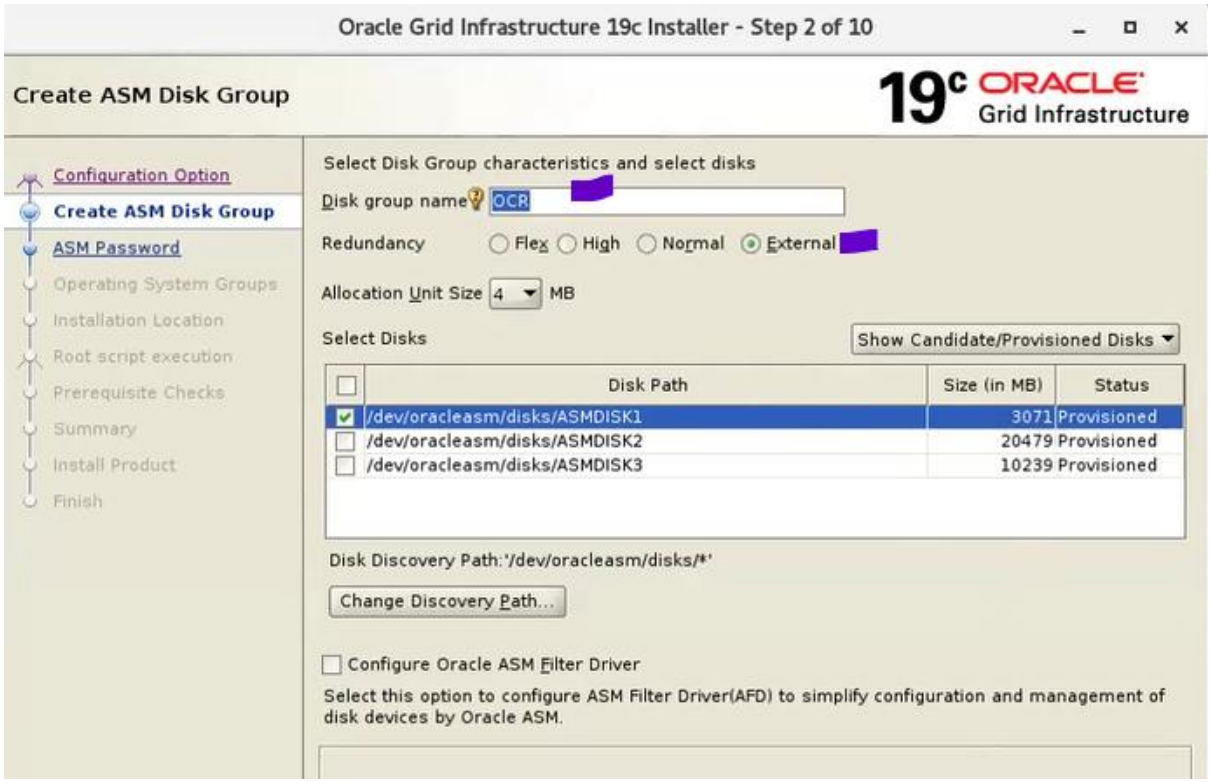


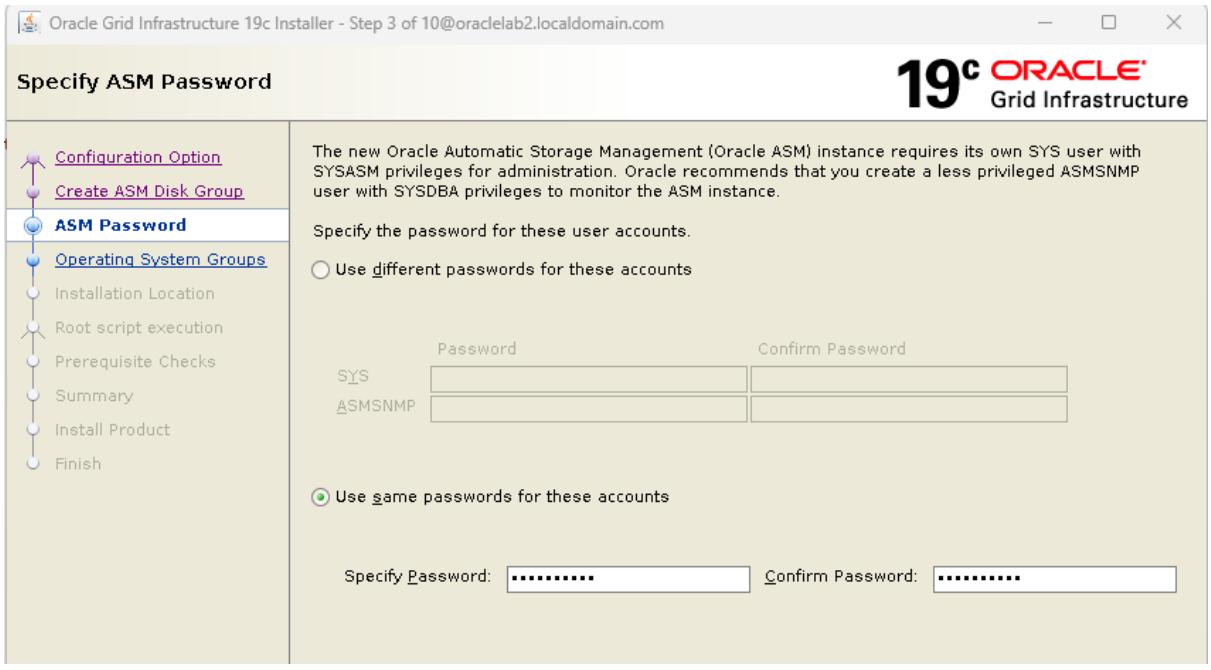


Sdb, etc.. Not able to recognize as those are virtual disks – asm understands only physical disks/raw disks, we need to make ASM fool by labelling

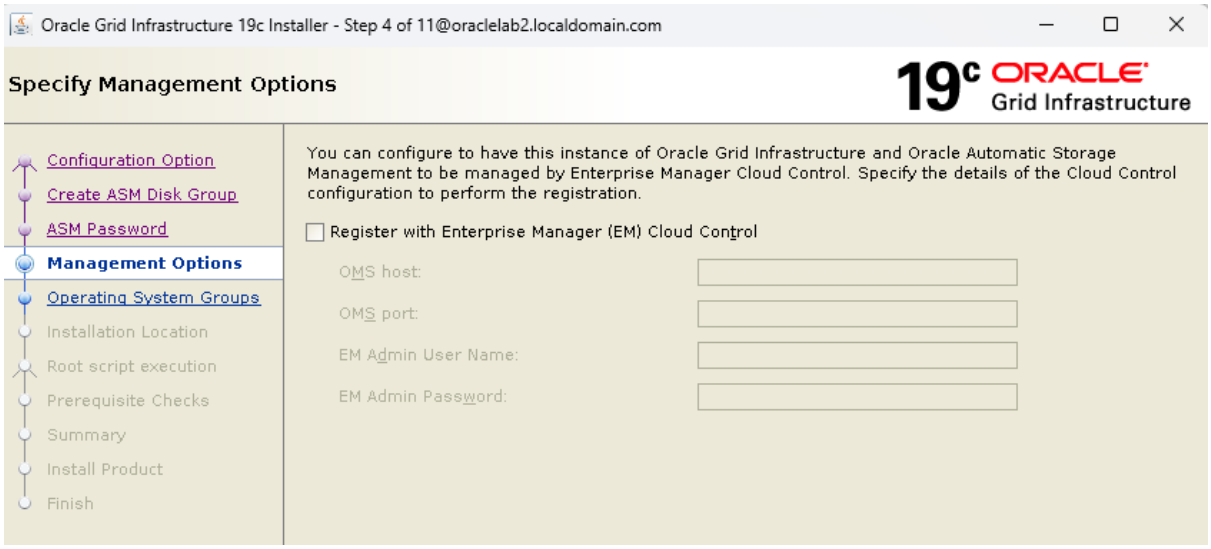
Change Directory path -

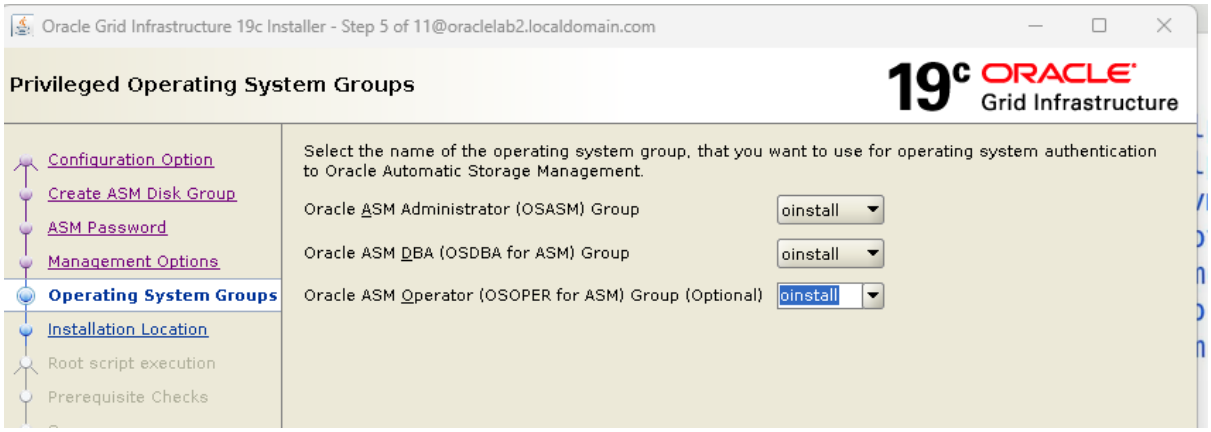


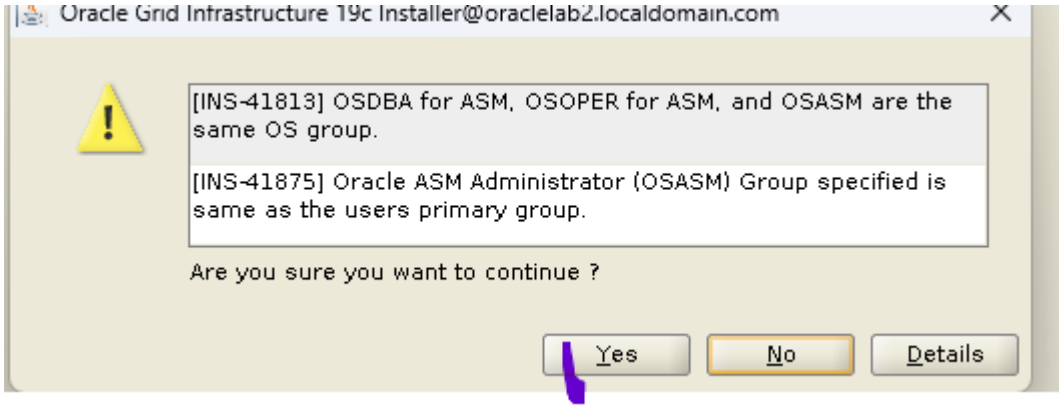

Oracle Grid Infrastructure 19c Installer - Step 2 of 10
Create ASM Disk Group
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Operating System Groups
Installation Location
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
Select Disk Group characteristics and select disks
Disk group name: OCR
Redundancy: Flex High Normal External
Allocation Unit Size: 4 MB
Select Disks
Show Candidate/Provisioned Disks
Disk Path
Size (in MB)
Status
/dev/oracleasm/disks/ASMDISK1 3071 Provisioned
/dev/oracleasm/disks/ASMDISK2 20479 Provisioned
/dev/oracleasm/disks/ASMDISK3 10239 Provisioned
Disk Discovery Path:'/dev/oracleasm/disks/*'
Change Discovery Path...
Configure Oracle ASM Filter Driver
Select this option to configure ASM Filter Driver(AFD) to simplify configuration and management of disk devices by Oracle ASM.

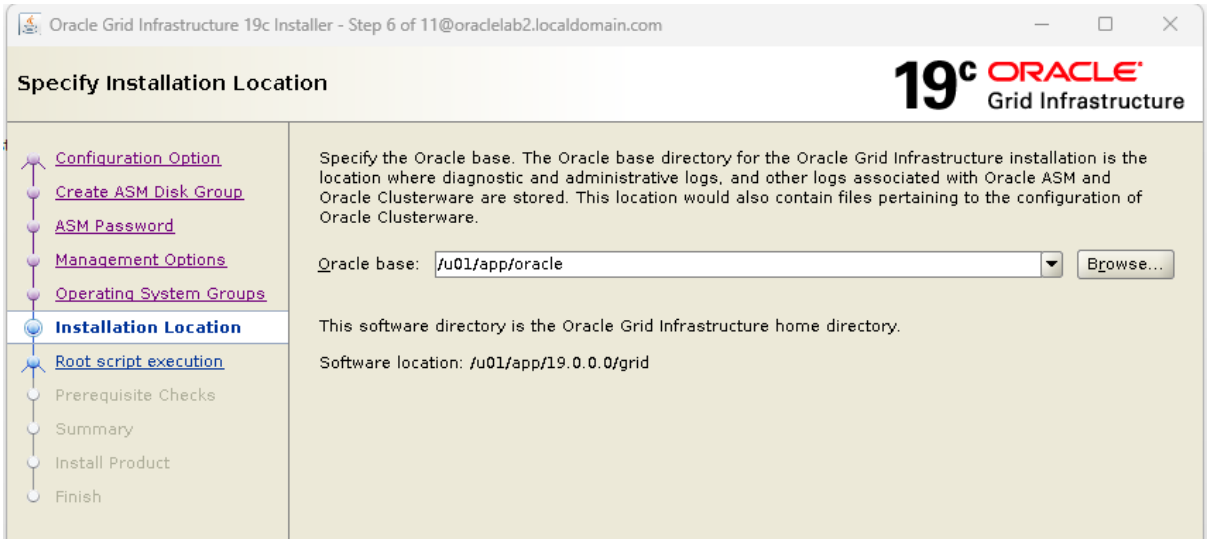

Oracle Grid Infrastructure 19c Installer - Step 3 of 10@oraclelab2.localdomain.com
Specify ASM Password
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Operating System Groups
Installation Location
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
The new Oracle Automatic Storage Management (Oracle ASM) instance requires its own SYS user with SYSASM privileges for administration. Oracle recommends that you create a less privileged ASMSNMP user with SYSDBA privileges to monitor the ASM instance.
Specify the password for these user accounts.
Use different passwords for these accounts
Password
Confirm Password
SYS
ASMSNMP
Use same passwords for these accounts
Specify Password:
Confirm Password:

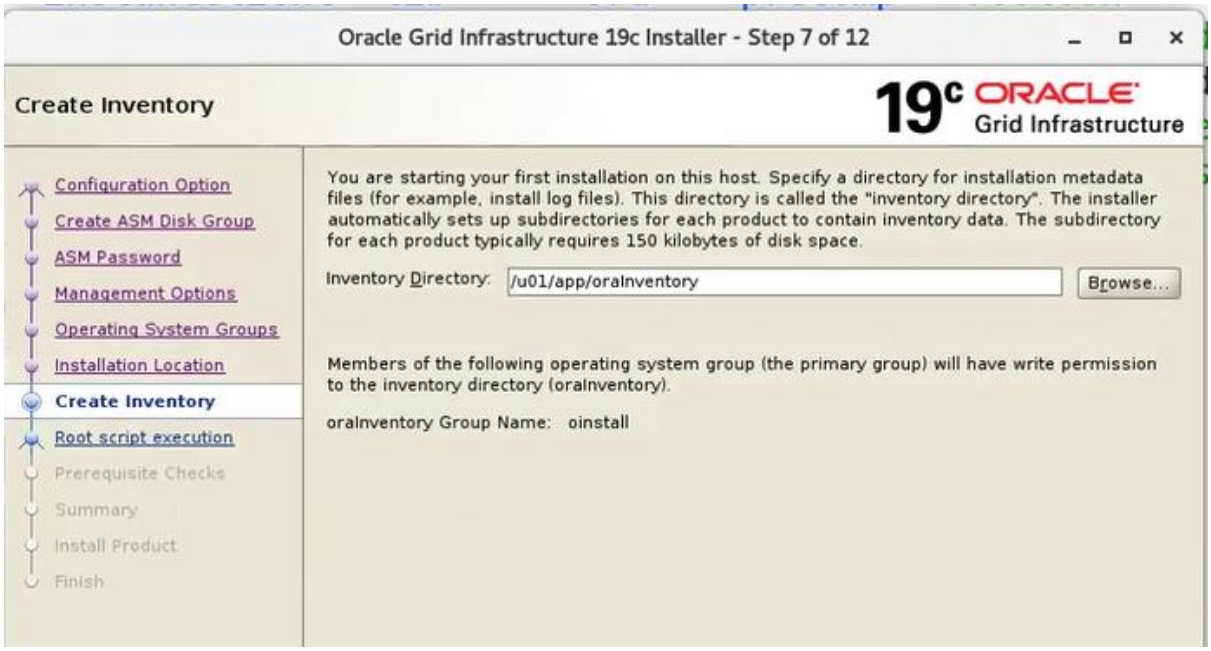
Mallik@123

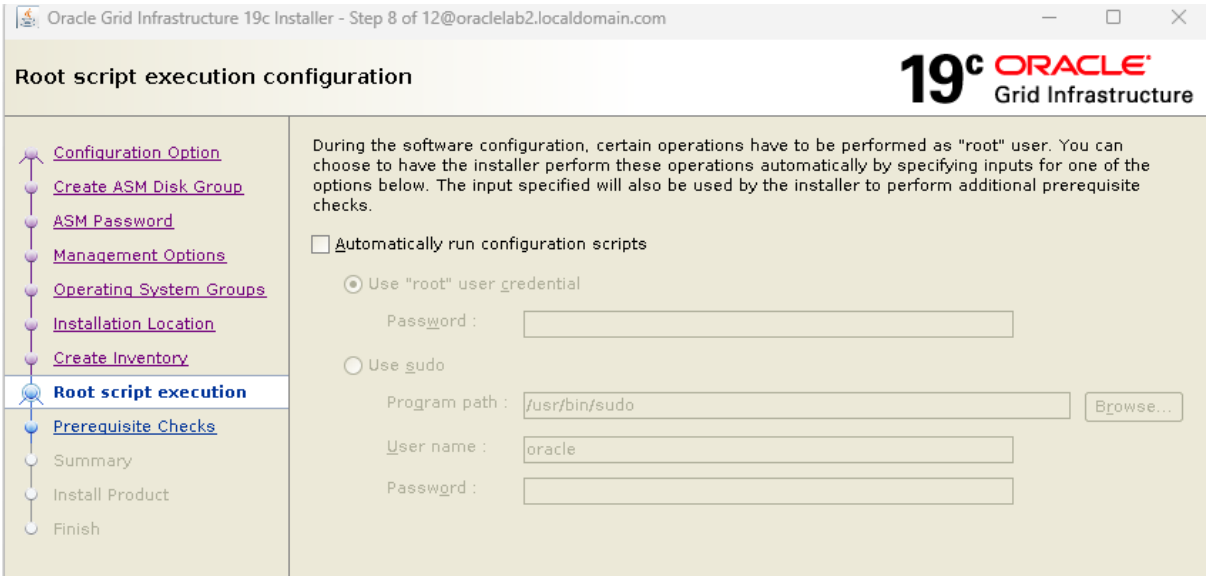

Oracle Grid Infrastructure 19c Installer - Step 4 of 11@oraclelab2.localdomain.com
Specify Management Options
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Management Options
Operating System Groups
Installation Location
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
You can configure to have this instance of Oracle Grid Infrastructure and Oracle Automatic Storage Management to be managed by Enterprise Manager Cloud Control. Specify the details of the Cloud Control configuration to perform the registration.
Register with Enterprise Manager (EM) Cloud Control
OMS host:
OMS port:
EM Admin User Name:
EM Admin Password:

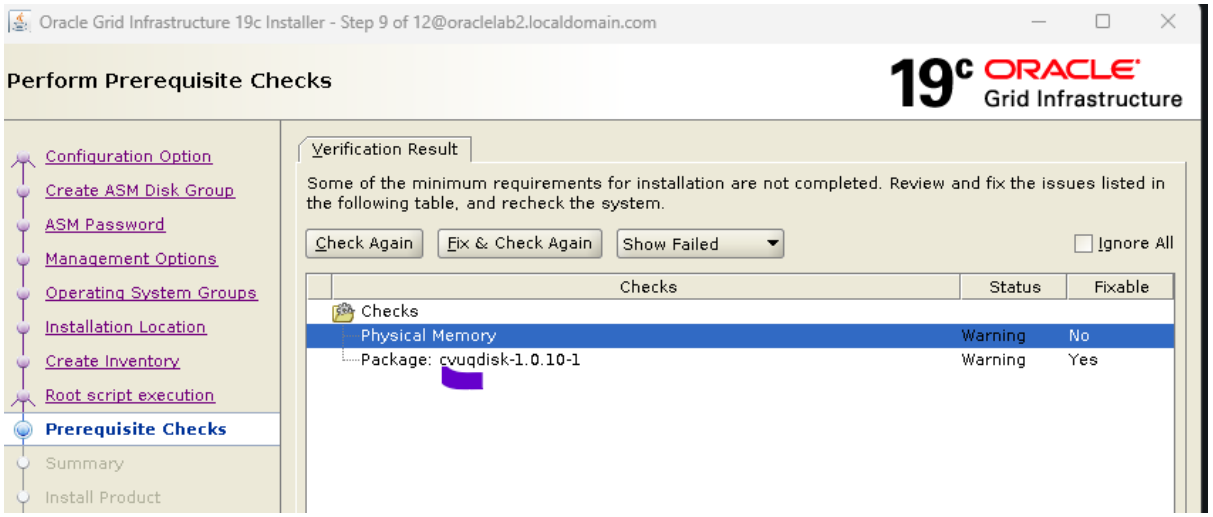

Oracle Grid Infrastructure 19c Installer - Step 5 of 11@oraclelab2.localdomain.com
Privileged Operating System Groups
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Management Options
Operating System Groups
Installation Location
Root script execution
Prerequisite Checks
Select the name of the operating system group, that you want to use for operating system authentication to Oracle Automatic Storage Management.
Oracle ASM Administrator (OSASM) Group
oinstall
Oracle ASM DBA (OSDBA for ASM) Group
oinstall
Oracle ASM Operator (OSOPER for ASM) Group (Optional)
oinstall


Oracle Grid Infrastructure 19c Installer@oraclelab2.localdomain.com
[INS-41813] OSDBA for ASM, OSOPER for ASM, and OSASM are the same OS group.
[INS-41875] Oracle ASM Administrator (OSASM) Group specified is same as the users primary group.
Are you sure you want to continue ?
Yes
No
Details


Oracle Grid Infrastructure 19c Installer - Step 6 of 11@oraclelab2.localdomain.com
Specify Installation Location
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Management Options
Operating System Groups
Installation Location
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
Specify the Oracle base. The Oracle base directory for the Oracle Grid Infrastructure installation is the location where diagnostic and administrative logs, and other logs associated with Oracle ASM and Oracle Clusterware are stored. This location would also contain files pertaining to the configuration of Oracle Clusterware.
Oracle base: /u01/app/oracle
Browse...
This software directory is the Oracle Grid Infrastructure home directory.
Software location: /u01/app/19.0.0.0/grid


Oracle Grid Infrastructure 19c Installer - Step 7 of 12
Create Inventory
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Management Options
Operating System Groups
Installation Location
Create Inventory
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
You are starting your first installation on this host. Specify a directory for installation metadata files (for example, install log files). This directory is called the "inventory directory". The installer automatically sets up subdirectories for each product to contain inventory data. The subdirectory for each product typically requires 150 kilobytes of disk space.
Inventory Directory: /u01/app/oraInventory
Browse...
Members of the following operating system group (the primary group) will have write permission to the inventory directory (oraInventory).
oraInventory Group Name: oinstall

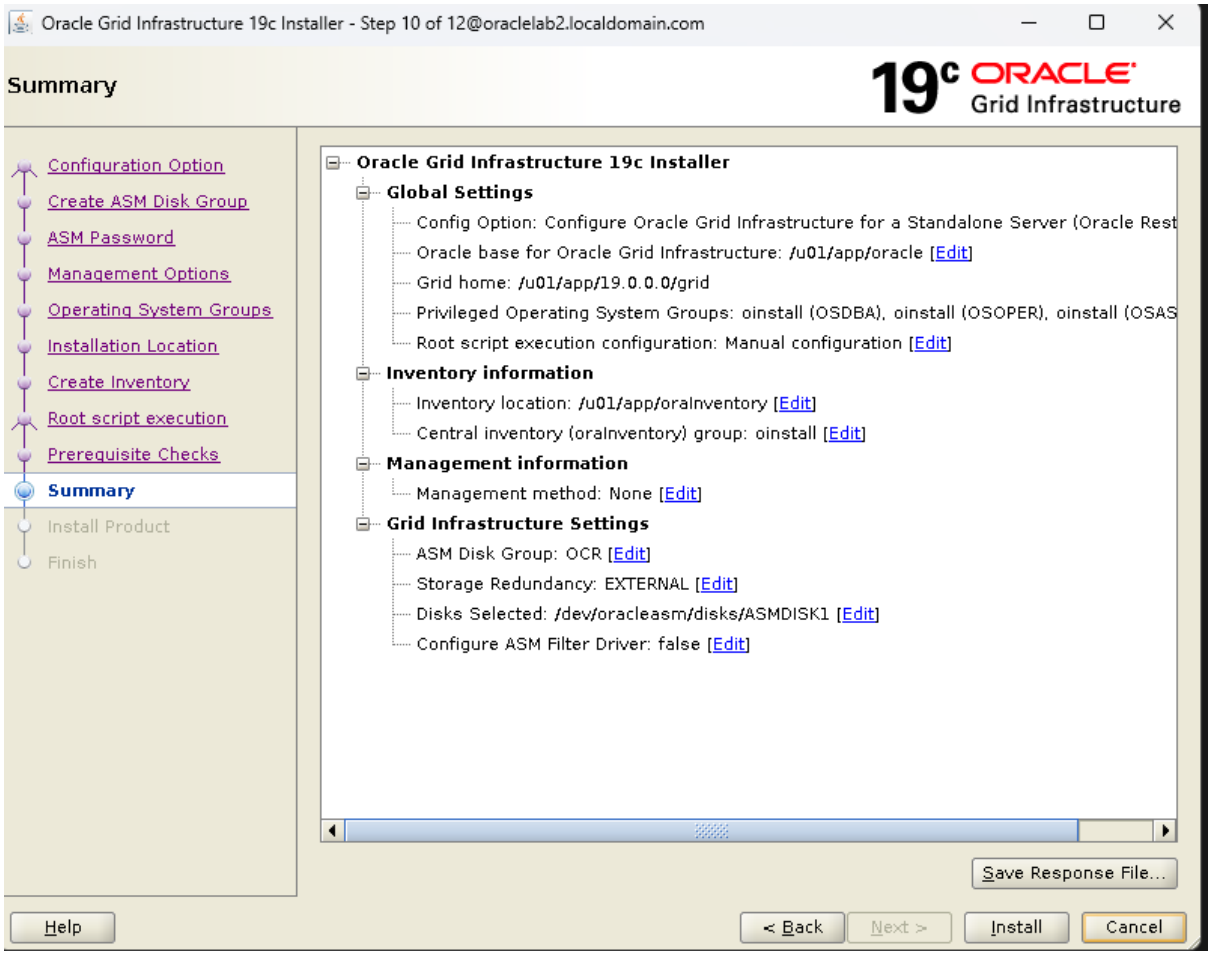

Oracle Grid Infrastructure 19c Installer - Step 8 of 12@oraclelab2.localdomain.com
Root script execution configuration
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Management Options
Operating System Groups
Installation Location
Create Inventory
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.
Automatically run configuration scripts
Use "root" user credential
Password :
Use sudo
Program path : /usr/bin/sudo
Browse...
User name : oracle
Password :

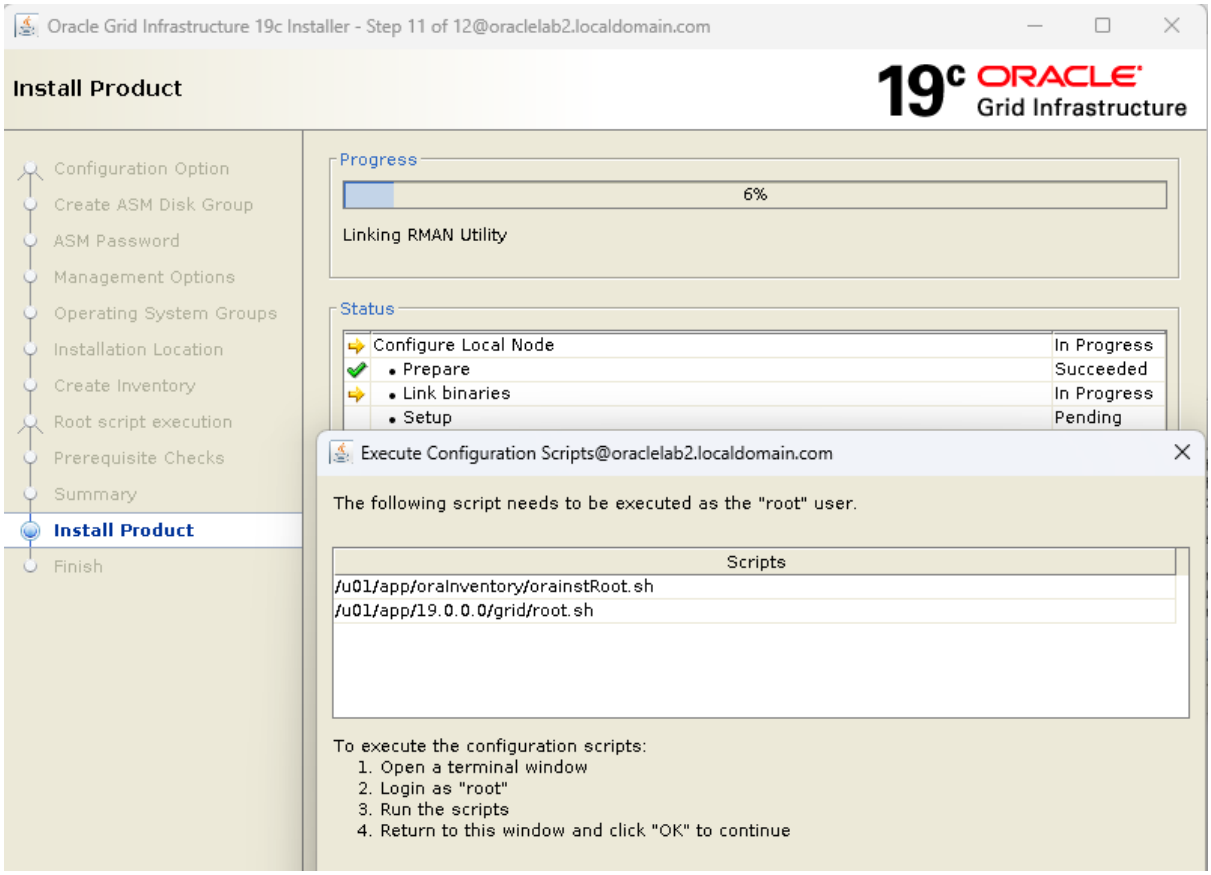


For this warning –

```
[root@oraclelab2 ~]# cd /u01/app/19.0.0.0/grid/cv/rpm/
[root@oraclelab2 rpm]# ll
total 12
-rw-r--r--. 1 oracle oinstall 11412 Mar 13  2019 cvuqdisk-1.0.10-1.rpm
[root@oraclelab2 rpm]# rpm -i cvuqdisk-1.0.10-1.rpm
Using default group oinstall to install package
[root@oraclelab2 rpm]#
```

Now, refresh – check again


Oracle Grid Infrastructure 19c Installer - Step 10 of 12@oraclelab2.localdomain.com
Summary
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Management Options
Operating System Groups
Installation Location
Create Inventory
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
Oracle Grid Infrastructure 19c Installer
Global Settings
Config Option: Configure Oracle Grid Infrastructure for a Standalone Server (Oracle Rest
Oracle base for Oracle Grid Infrastructure: /u01/app/oracle [Edit]
Grid home: /u01/app/19.0.0.0/grid
Privileged Operating System Groups: oinstall (OSDBA), oinstall (OSOPER), oinstall (OSAS
Root script execution configuration: Manual configuration [Edit]
Inventory information
Inventory location: /u01/app/oraInventory [Edit]
Central inventory (oraInventory) group: oinstall [Edit]
Management information
Management method: None [Edit]
Grid Infrastructure Settings
ASM Disk Group: OCR [Edit]
Storage Redundancy: EXTERNAL [Edit]
Disks Selected: /dev/oracleasm/disks/ASMDISK1 [Edit]
Configure ASM Filter Driver: false [Edit]
Save Response File...
Help
< Back
Next >
Install
Cancel


Oracle Grid Infrastructure 19c Installer - Step 11 of 12@oraclelab2.localdomain.com
Install Product
19c ORACLE Grid Infrastructure
Configuration Option
Create ASM Disk Group
ASM Password
Management Options
Operating System Groups
Installation Location
Create Inventory
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
Progress
6%
Linking RMAN Utility
Status
Configure Local Node
In Progress
Prepare
Succeeded
Link binaries
In Progress
Setup
Pending
Execute Configuration Scripts@oraclelab2.localdomain.com
The following script needs to be executed as the "root" user.
Scripts
/u01/app/oraInventory/orainstRoot.sh
/u01/app/19.0.0.0/grid/root.sh
To execute the configuration scripts:
1. Open a terminal window
2. Login as "root"
3. Run the scripts
4. Return to this window and click "OK" to continue

```
Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Using configuration parameter file: /u01/app/19.0.0.0/grid/crs/install/crsconfig_params
The log of current session can be found at:
  /u01/app/oracle/crsdata/oraclelab2/crsconfig/roothas_2026-10-11_04-47-47PM.log
LOCAL ADD MODE
Creating OCR keys for user 'oracle', privgrp 'oinstall'..
Operation successful.
LOCAL ONLY MODE
Successfully accumulated necessary OCR keys.
Creating OCR keys for user 'root', privgrp 'root'..
Operation successful.
CRS-4664: Node oraclelab2 successfully pinned.
2026/10/11 16:48:21 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'

oraclelab2     2026/10/11 16:49:53     /u01/app/oracle/crsdata/oraclelab2/olr/backup_20261011_164953.olr     724960844
2026/10/11 16:49:56 CLSRSC-327: Successfully configured Oracle Restart for a standalone server
[root@oraclelab2 ~]#
```



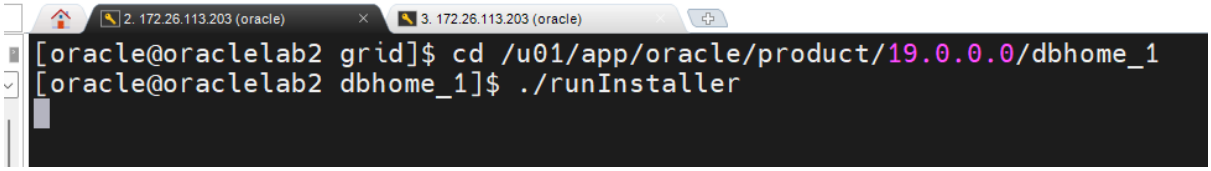
********** done with ASM configuration **

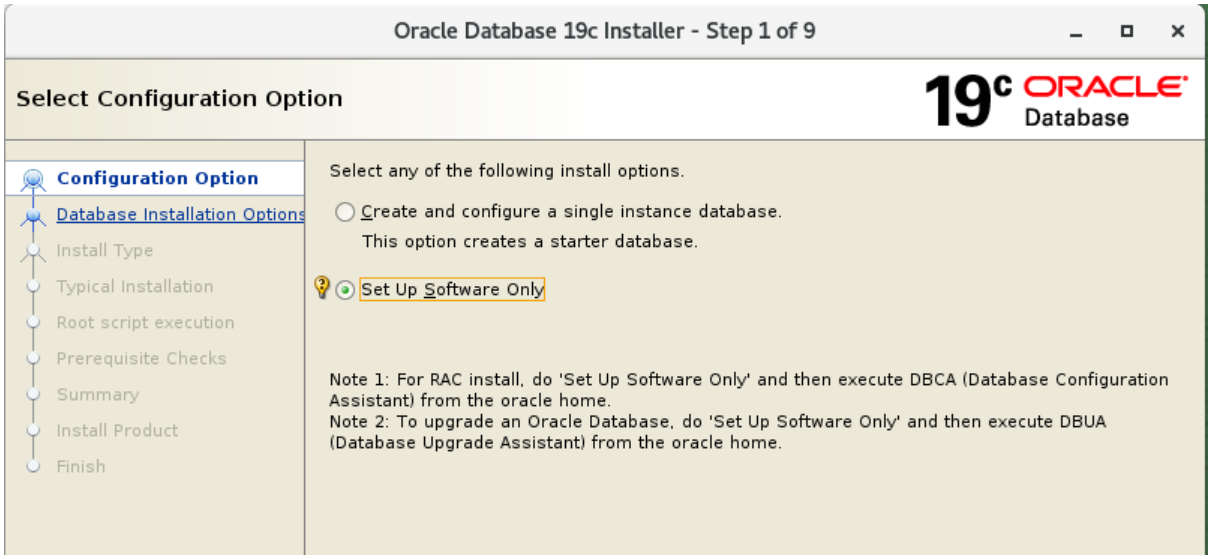
3. Database SW Installation / Oracle SW Installation / Oracle Home Installation (from GUI screen as oracle user)

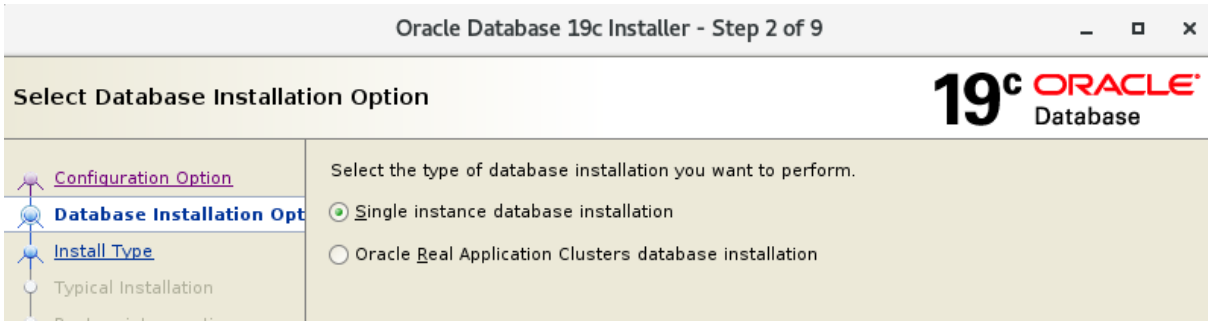
Login with oracle user

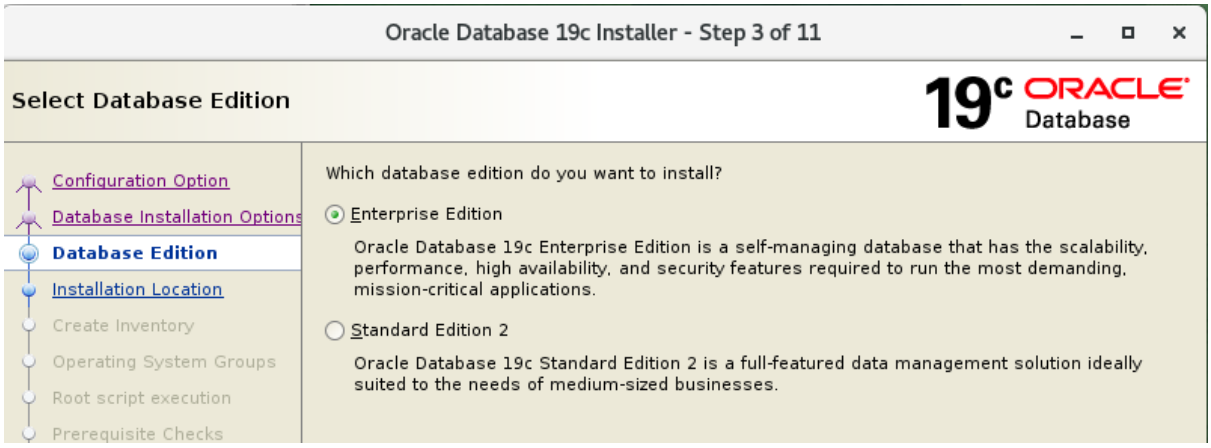
$ cd /u01/app/oracle/product/19.0.0.0/dbhome_1

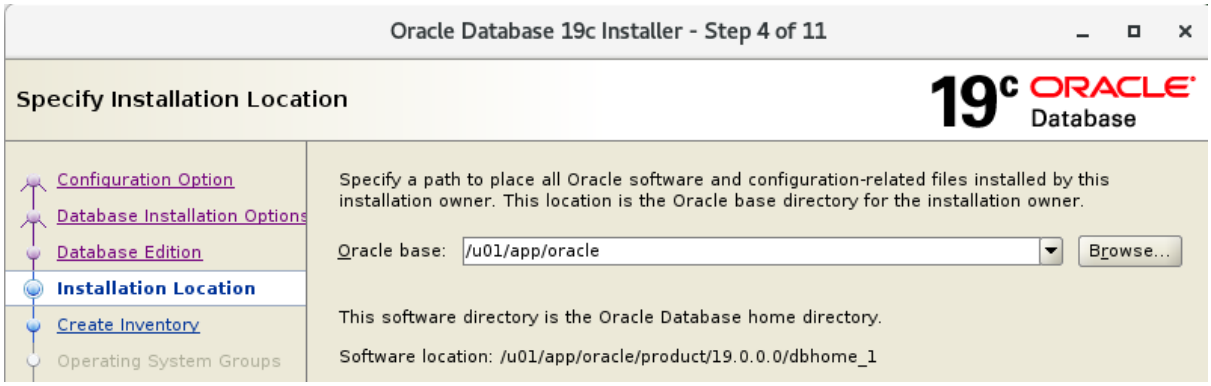
$ ./runInstaller

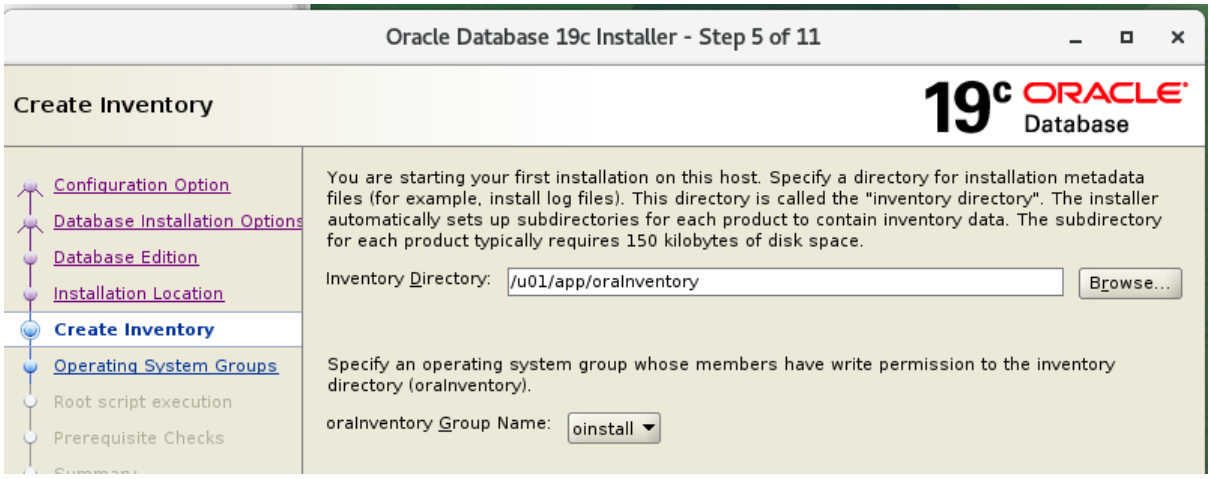


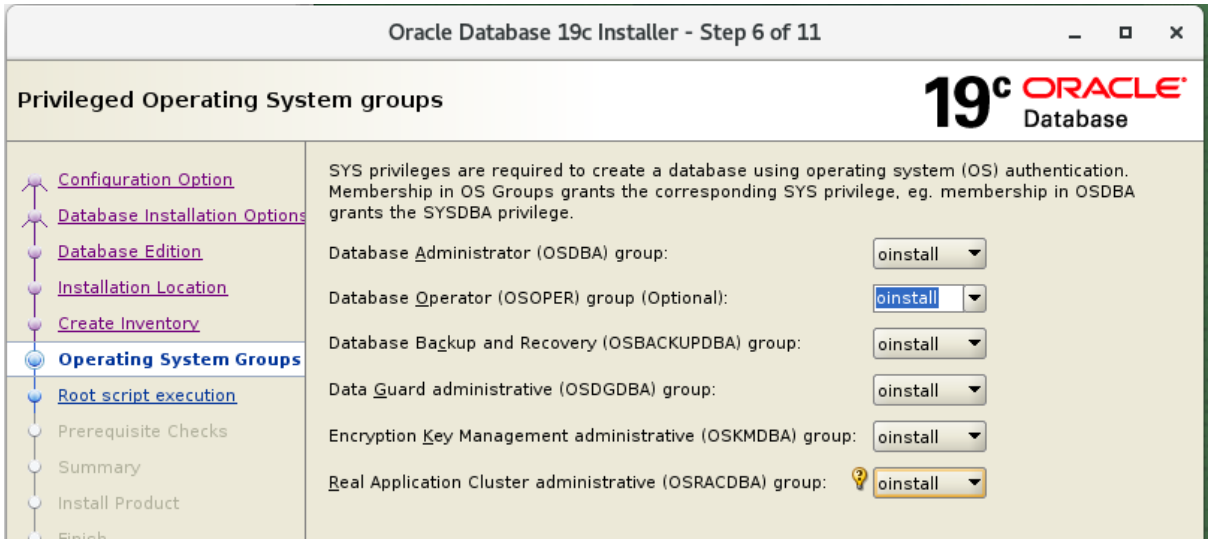


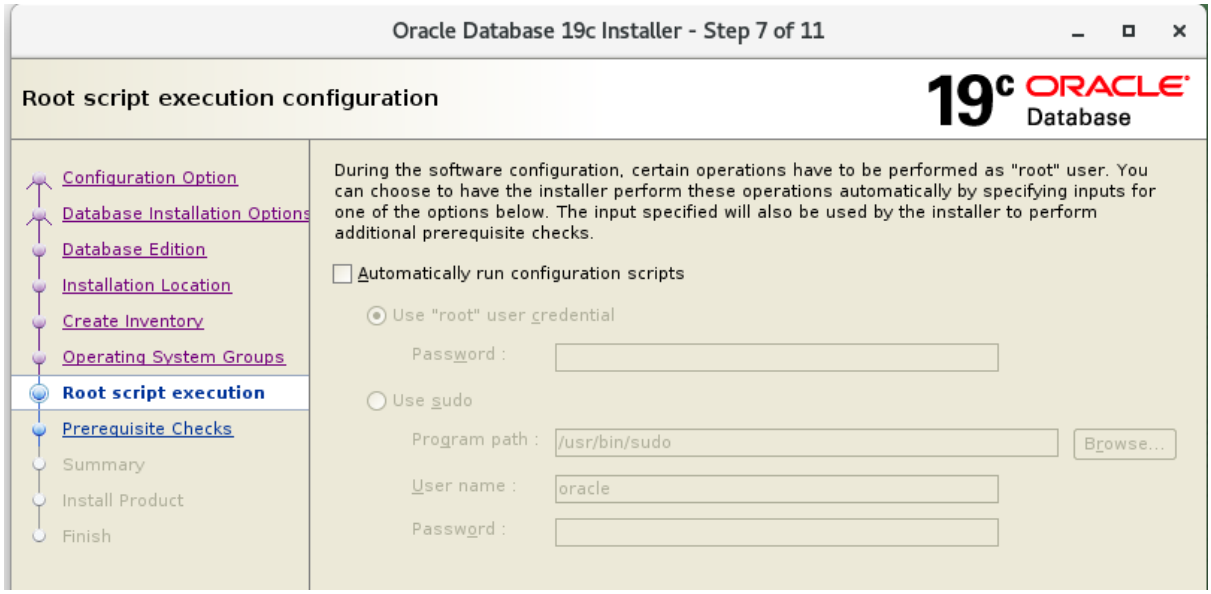


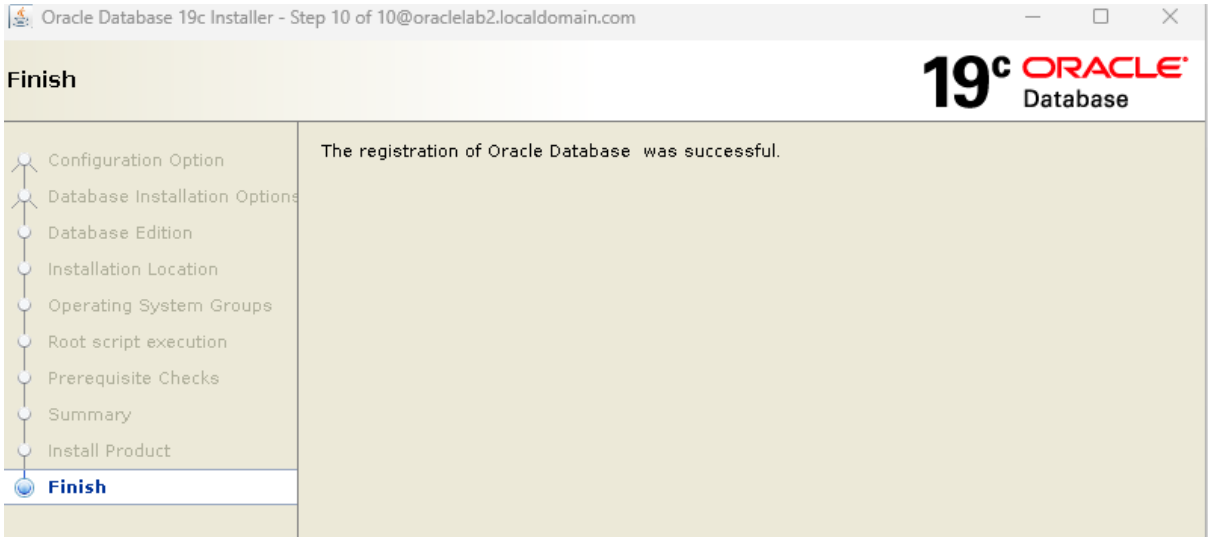

Oracle Database 19c Installer - Step 3 of 11
Select Database Edition
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
Which database edition do you want to install?
Enterprise Edition
Oracle Database 19c Enterprise Edition is a self-managing database that has the scalability, performance, high availability, and security features required to run the most demanding, mission-critical applications.
Standard Edition 2
Oracle Database 19c Standard Edition 2 is a full-featured data management solution ideally suited to the needs of medium-sized businesses.


Oracle Database 19c Installer - Step 4 of 11
Specify Installation Location
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Specify a path to place all Oracle software and configuration-related files installed by this installation owner. This location is the Oracle base directory for the installation owner.
Oracle base: /u01/app/oracle
Browse...
This software directory is the Oracle Database home directory.
Software location: /u01/app/oracle/product/19.0.0.0/dbhome_1


Oracle Database 19c Installer - Step 5 of 11
Create Inventory
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
You are starting your first installation on this host. Specify a directory for installation metadata files (for example, install log files). This directory is called the "inventory directory". The installer automatically sets up subdirectories for each product to contain inventory data. The subdirectory for each product typically requires 150 kilobytes of disk space.
Inventory Directory: /u01/app/oraInventory
Browse...
Specify an operating system group whose members have write permission to the inventory directory (oraInventory).
oraInventory Group Name: oinstall







*** Oracle HOME installation is completed ***

4. Create Database -> TESTDB with oracle user

First, before creating the db, add asm storage

```
Add asm storage and create ASM diskgroups (+DATA & +RECO)

Add virtual disk to oraclelab2 (20G & 10G)
/dev/sdc - 20G (Virtual disk)
/dev/sdd - 10G (Virtual disk)

OS level make partition to virtual
fdisk /dev/sdc
fdisk /dev/sdd
    - n
    - w
/dev/sdc - /dev/sdc1 - 20G
/dev/sdd - /dev/sdd1 - 10G

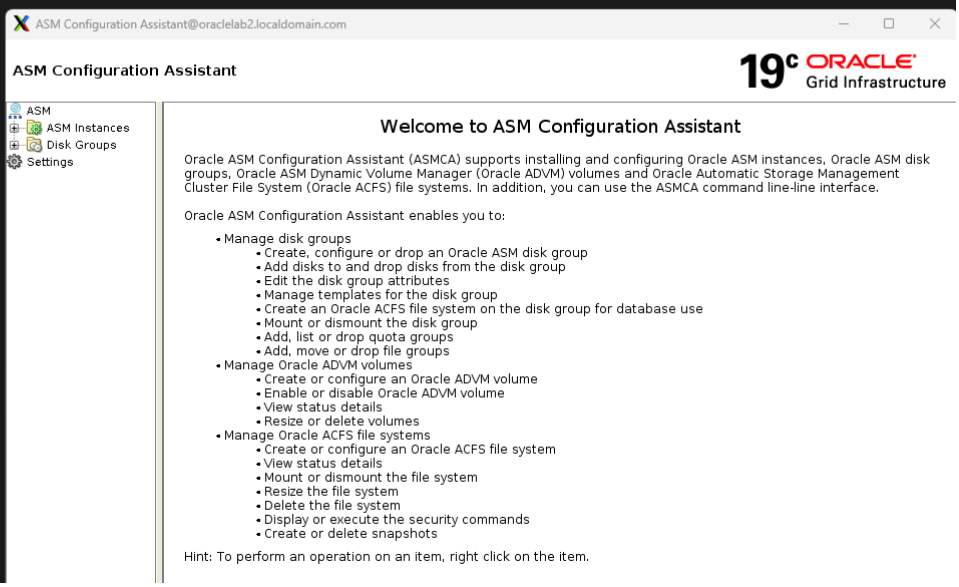
oracleasm createdisk ASMDISK2 /dev/sdc1 --- DATA/DBDATA
oracleasm createdisk ASMDISK3 /dev/sdd1 --- RECO/FRA/ARCHIVE

- cd /u01/app/19.0.0.0/grid/bin (from GUI screen as oracle user)
- ./asmca (+DATA & +RECO)

    $ cd /u01/app/oracle/product/19.0.0.0/dbhome_1/bin (from GUI screen as oracle user)
    $ ./dbca
```

asmca – asm configuration assistant

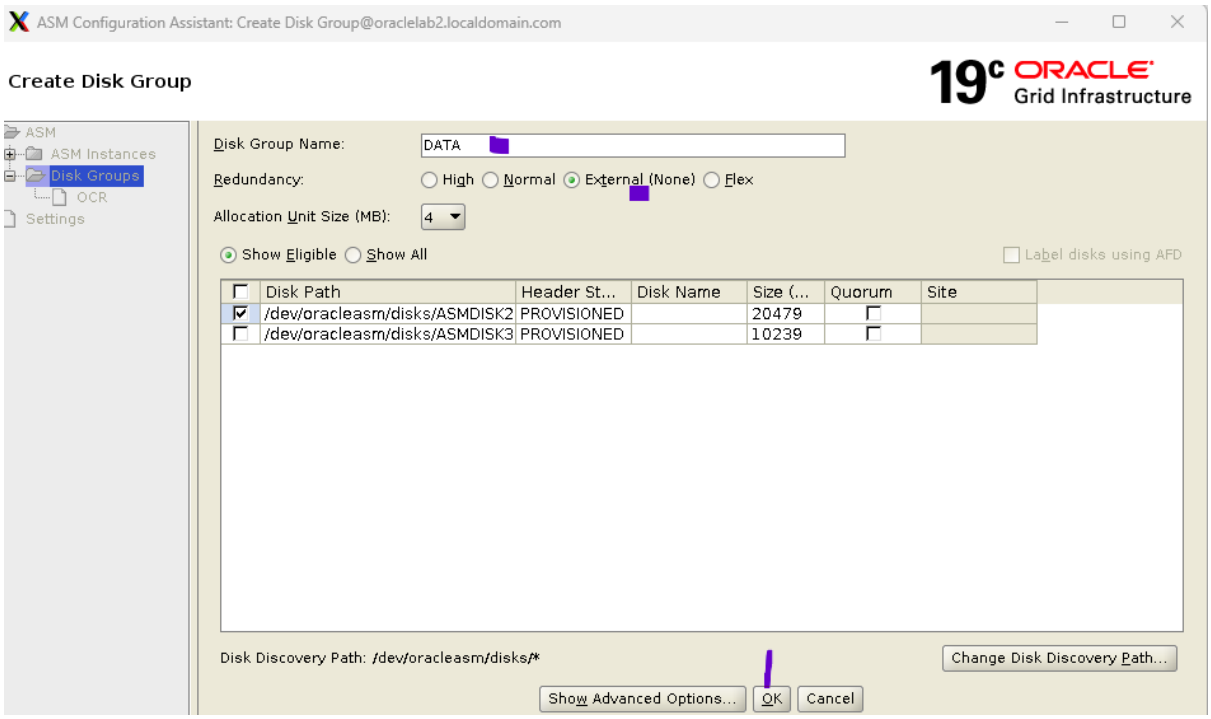
```
[oracle@oraclelab2 bin]$ cd /u01/app/19.0.0.0/grid/bin/
[oracle@oraclelab2 bin]$ ll asmca
-rwxr-x---. 1 root oinstall 7912 Oct 11 16:44 asmca
[oracle@oraclelab2 bin]$ ./asmca
```

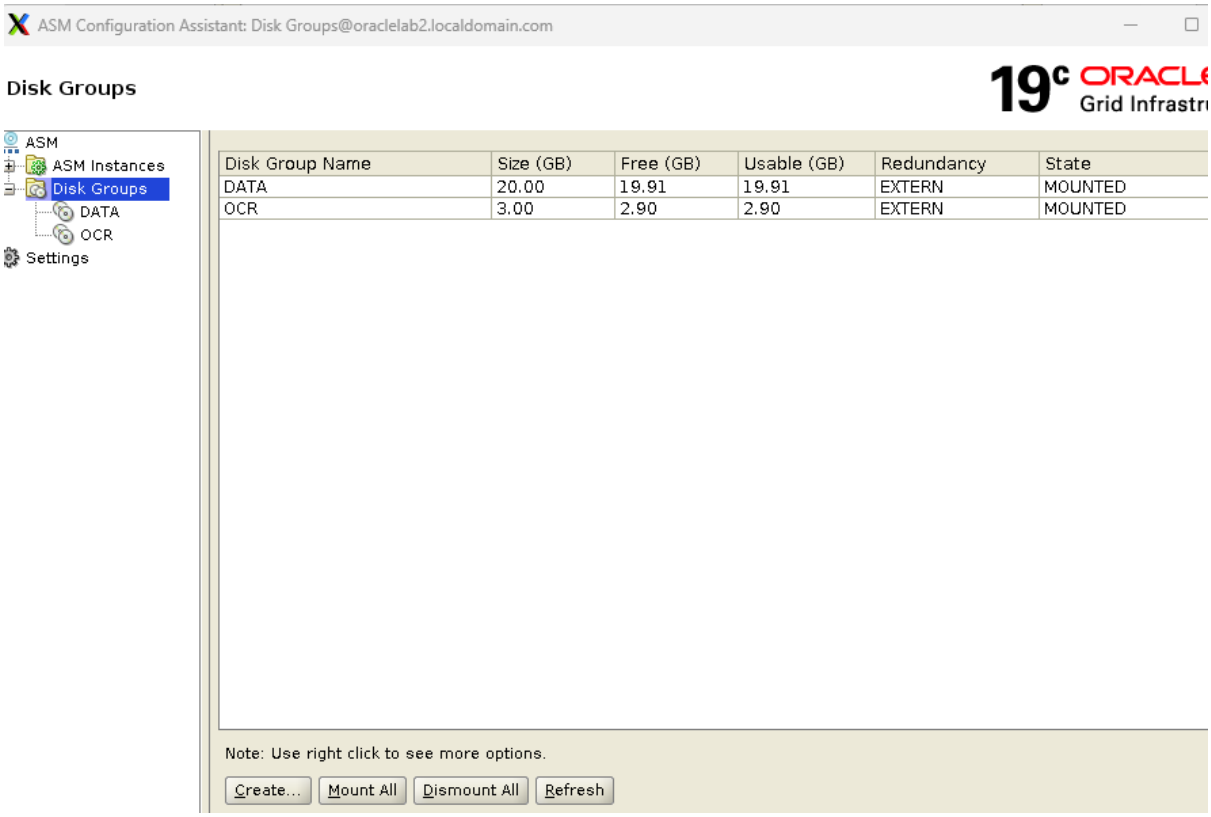


Diskgroups

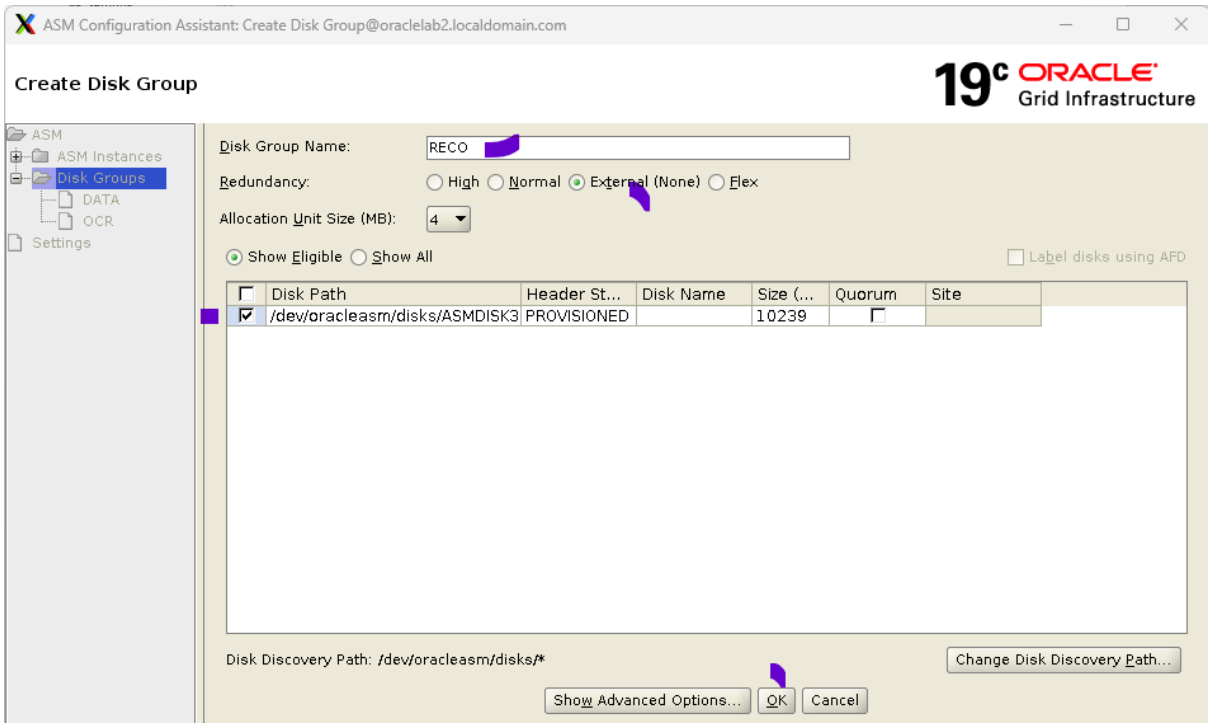

ASM Configuration Assistant: Disk Groups@oraclelab2.localdomain.com
Disk Groups
19c ORACLE Grid Infrastructure
ASM
ASM Instances
Disk Groups
OCR
Settings
Disk Group Name | Size (GB) | Free (GB) | Usable (GB) | Redundancy | State
OCR | 3.00 | 2.90 | 2.90 | EXTERN | MOUNTED
Note: Use right click to see more options.
Create... Mount All Dismount All Refresh

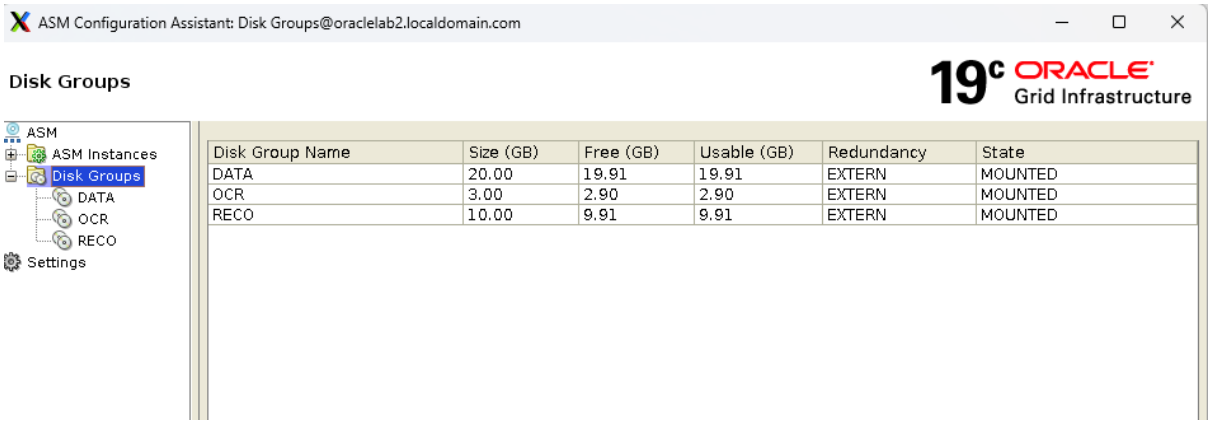
CREATE


ASM Configuration Assistant: Create Disk Group@oraclelab2.localdomain.com
Create Disk Group
19c ORACLE Grid Infrastructure
ASM
ASM Instances
Disk Groups
OCR
Settings
Disk Group Name: DATA
Redundancy: High Normal External (None) Flex
Allocation Unit Size (MB): 4
Show Eligible Show All
Label disks using AFD
Disk Path | Header St... | Disk Name | Size (... | Quorum | Site
/dev/oracleasm/disks/ASMDISK2 | PROVISIONED | | 20479
/dev/oracleasm/disks/ASMDISK3 | PROVISIONED | | 10239
Disk Discovery Path: /dev/oracleasm/disks/*
Change Disk Discovery Path...
Show Advanced Options... OK Cancel


ASM Configuration Assistant: Disk Groups@oraclelab2.localdomain.com
Disk Groups
19c ORACLE
Grid Infrastr
ASM
ASM Instances
Disk Groups
DATA
OCR
Settings
Disk Group Name | Size (GB) | Free (GB) | Usable (GB) | Redundancy | State
DATA | 20.00 | 19.91 | 19.91 | EXTERN | MOUNTED
OCR | 3.00 | 2.90 | 2.90 | EXTERN | MOUNTED
Note: Use right click to see more options.
Create... Mount All Dismount All Refresh

CREATE ->

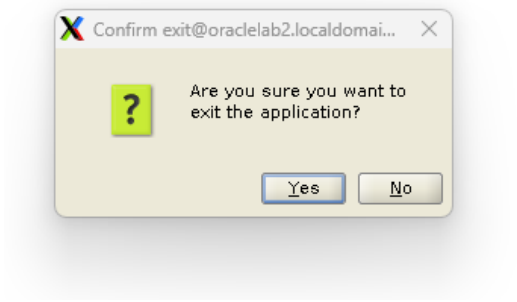

ASM Configuration Assistant: Create Disk Group@oraclelab2.localdomain.com
Create Disk Group
19c ORACLE
Grid Infrastructure
ASM
ASM Instances
Disk Groups
DATA
OCR
Settings
Disk Group Name: RECO
Redundancy: High Normal External (None) Flex
Allocation Unit Size (MB): 4
Show Eligible Show All
Label disks using AFD
Disk Path | Header St... | Disk Name | Size (... | Quorum | Site
/dev/oracleasm/disks/ASMDISK3 | PROVISIONED | | 10239 |
Disk Discovery Path: /dev/oracleasm/disks/*
Change Disk Discovery Path...
Show Advanced Options... OK Cancel



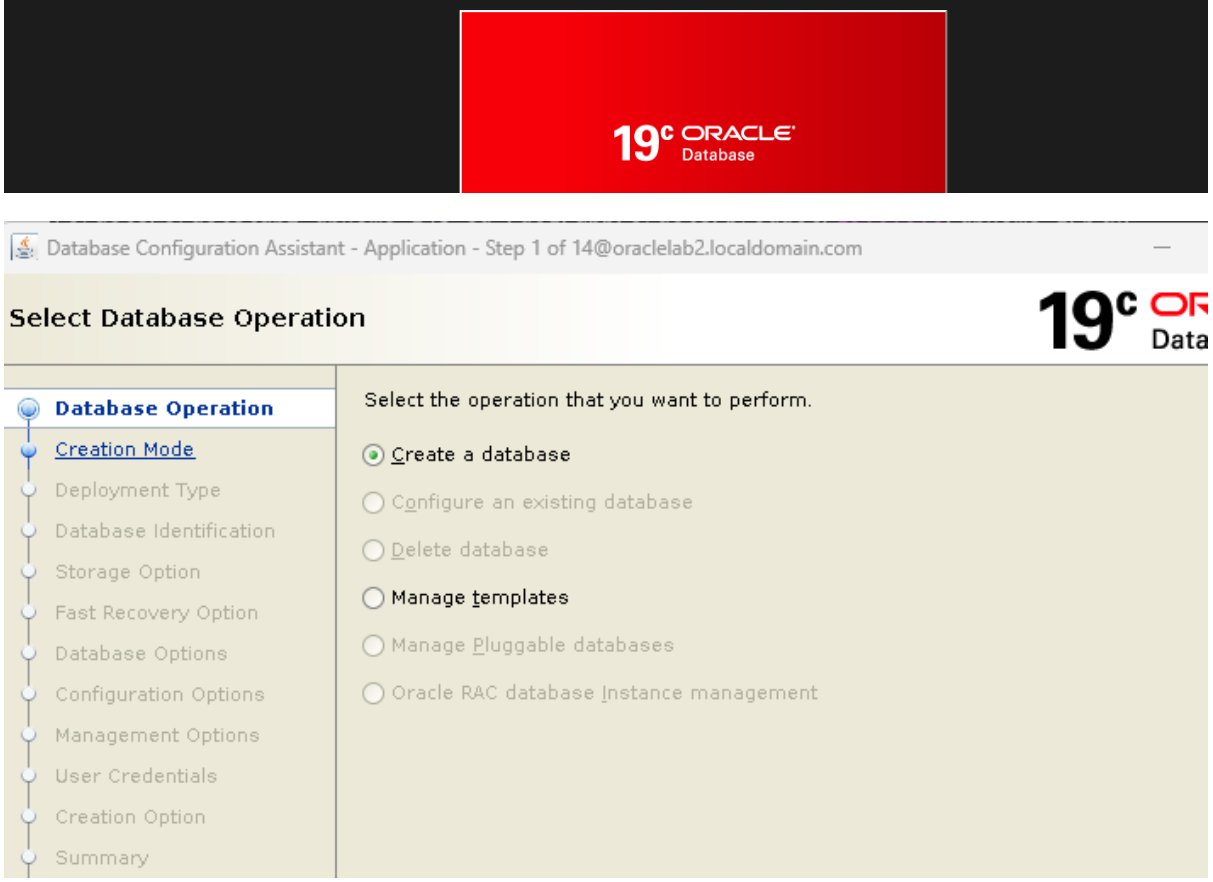
Verify -

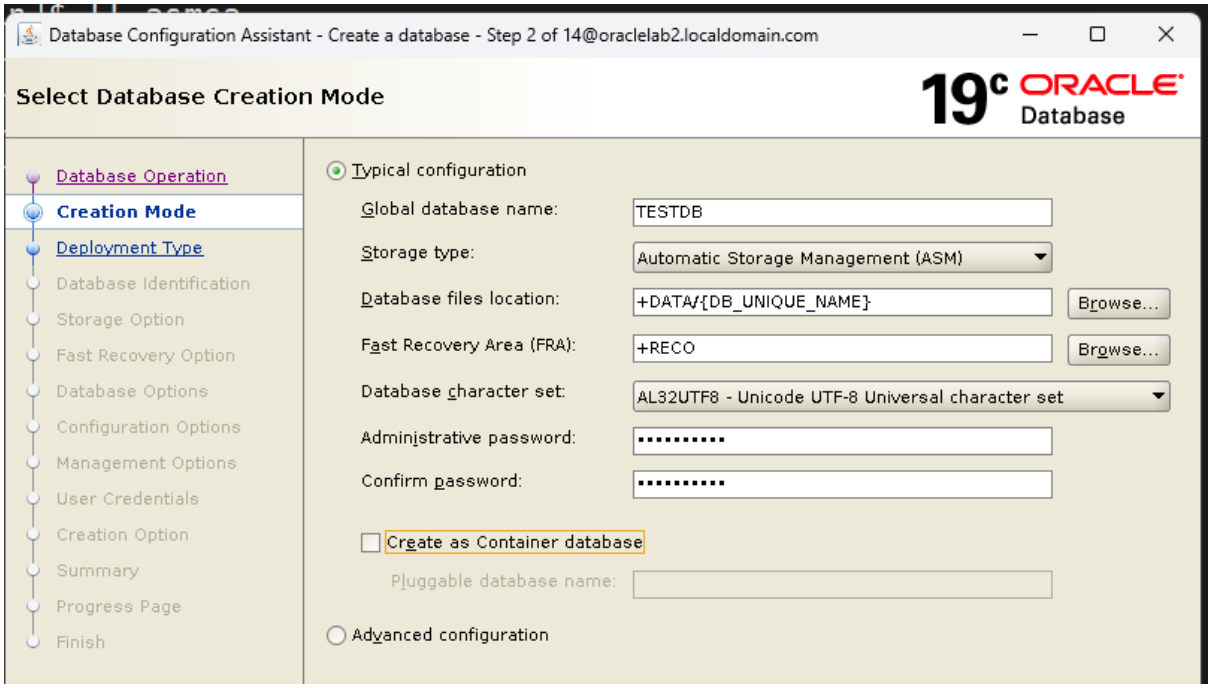
```
[oracle@oraclelab2 ~]$ lsdg
bash: lsdg: command not found...
[oracle@oraclelab2 ~]$ ps -ef | grep smon
oracle   22549     1  0 17:02 ?        00:00:00 asm_smon_+ASM
oracle   31499 31285  0 17:29 pts/2    00:00:00 grep --color=auto smon
[oracle@oraclelab2 ~]$ .oraenv
bash: .oraenv: command not found...
[oracle@oraclelab2 ~]$ . oraenv
ORACLE_SID = [oracle] ? +ASM
The Oracle base has been set to /u01/app/oracle
[oracle@oraclelab2 ~]$ asmcmd -p
ASMCMD [+] > lsdg
State    Type    Rebal  Sector  Logical_Sector  Block       AU  Total_MB  Free_MB  Req_mir_free_MB  Usable_file_MB  Offline
_disks  Voting_files  Name
MOUNTED  EXTERN  N         512             512   4096  4194304     20476    20388                0           20388
     0             N  DATA/
MOUNTED  EXTERN  N         512             512   4096  4194304      3068     2968                0            2968
     0             N  OCR/
MOUNTED  EXTERN  N         512             512   4096  4194304     10236    10148                0           10148
     0             N  RECO/
ASMCMD [+] >
```

EXIT - OK

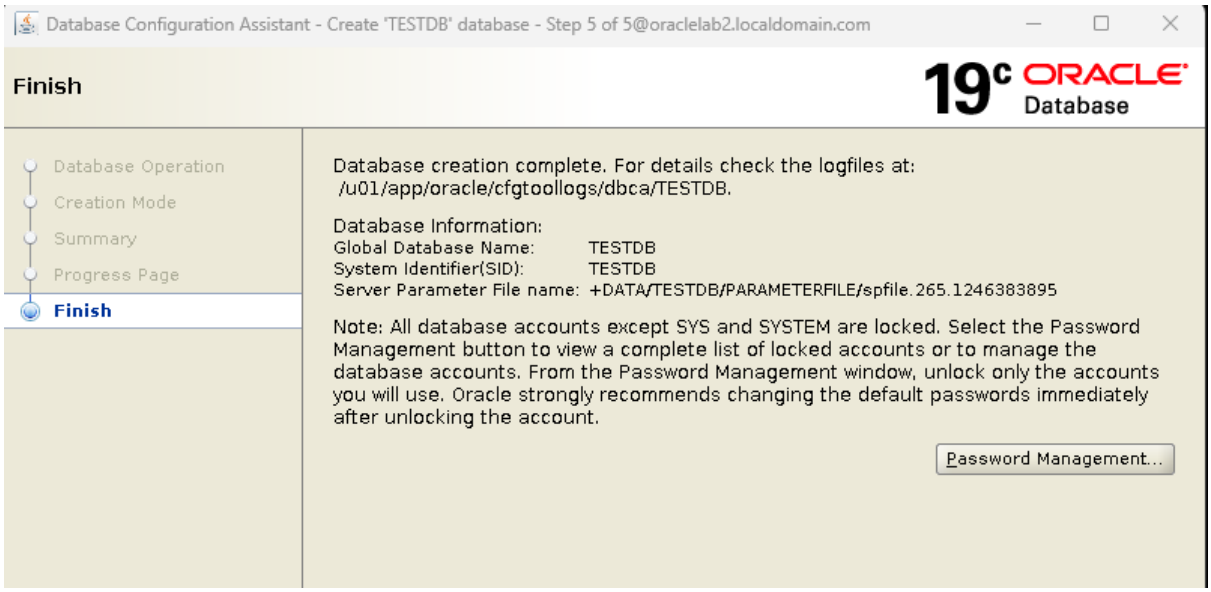


```
[oracle@oraclelab2 dbhome_1]$ cd /u01/app/oracle/product/19.0.0.0/dbhome_1/bin/
[oracle@oraclelab2 bin]$ ./dbca
```





Mallik@123




```
[oracle@oraclelab2 bin]$ ps -ef | grep smon
oracle    9820     1  0 17:47 ?        00:00:00 ora_smon_TESTDB
oracle   14067  3693  0 18:01 pts/1    00:00:00 grep --color=auto smon
oracle   22549     1  0 17:02 ?        00:00:00 asm_smon_+ASM
[oracle@oraclelab2 bin]$
```